Semiconductor Switching Devices

- **Power diodes**
  - Standard rectifier diode
  - Fast-recovery / ultrafast diode
  - Schottky diode
  - PIN diode (used where high voltage blocking is needed)
- **Transistors (used as switches)**
  - Bipolar Junction Transistor (BJT – power BJT)
  - MOSFET (power MOSFET)
  - IGBT (Insulated Gate Bipolar Transistor)
  - SIT / JFET / other FET variants (seen in some special SMPS/RF uses)
  - Darlington transistor (BJT pair for higher gain)
- **Thyristor family (PNPN devices)**
  - SCR (Silicon Controlled Rectifier)
  - TRIAC (for AC switching)
  - GTO (Gate Turn-Off thyristor)
  - IGCT (Integrated Gate Commutated Thyristor)
  - LASCR (Light-activated SCR)
  - RCT (Reverse Conducting Thyristor)
  - MCT (MOS-Controlled Thyristor)
  - SITH (Static Induction Thyristor)

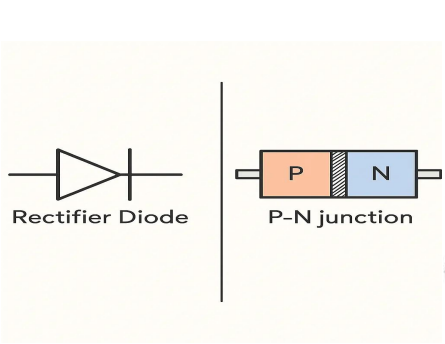
How they are often classified in power electronics

- By terminals:
  - Two-terminal: power **diodes**.
  - Three-terminal: **BJTs, MOSFETs, IGBTs, SCR (if you count gate)**.
  - Four-terminal or special: some optically or sensor-type power devices.
- By control capability:
  - Semi-controlled: SCR (controlled turn-on, natural/line commutation for turn-off).
  - Fully controlled: BJT, MOSFET, IGBT, GTO, IGCT, MCT (both turn-on and turn-off via gate/base).

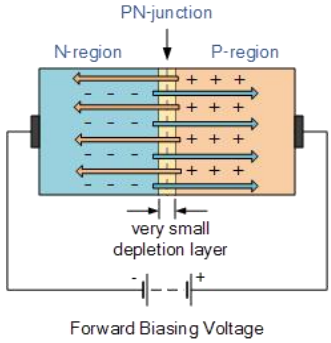
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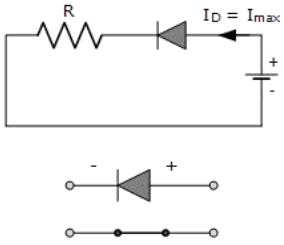
1. Standard rectifier diode

1. What is a Standard Rectifier Diode?







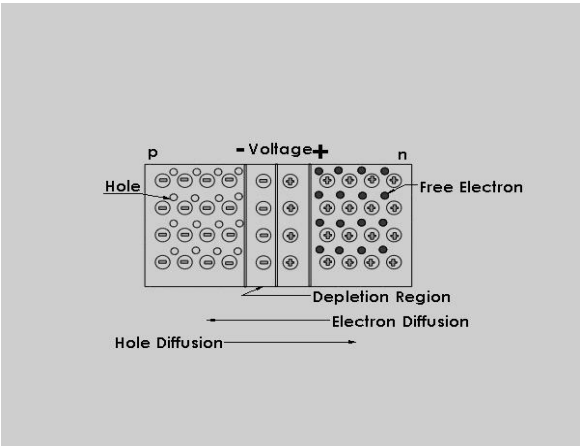


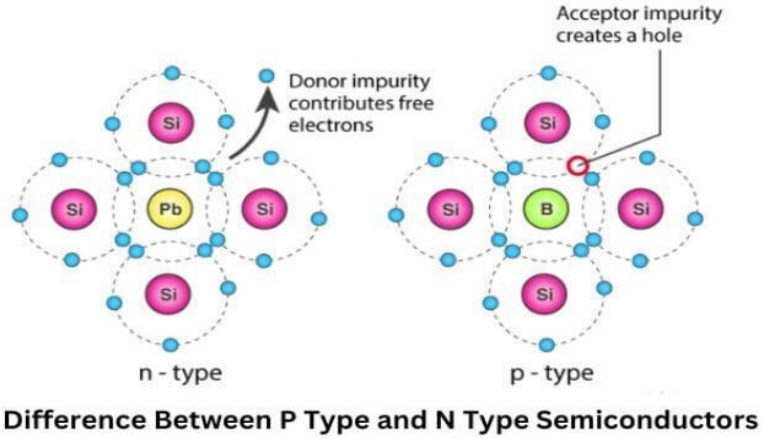
A **standard rectifier diode** is a **PN junction semiconductor device** that allows **current to flow in only one direction** and is mainly used to **convert AC into DC**.

Key identity points:

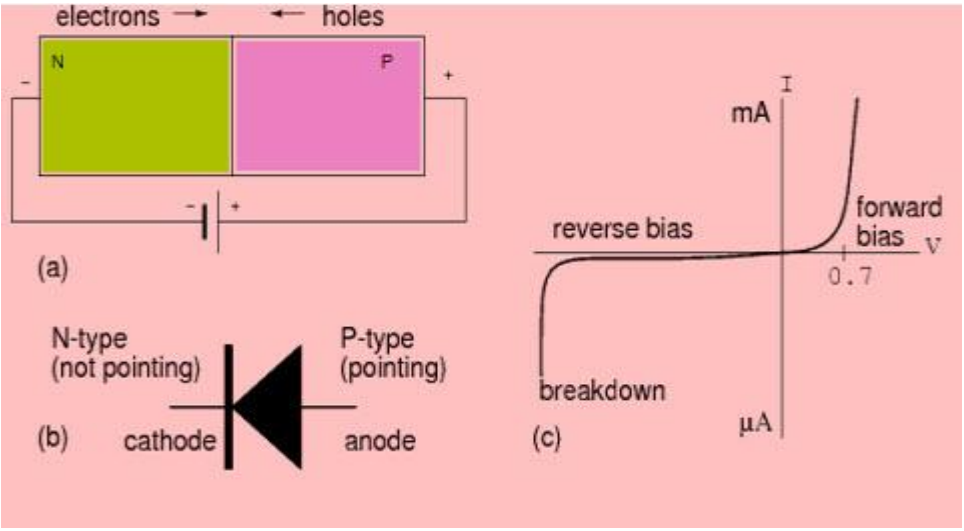
- Works on **PN junction principle**
- Used in **power supplies**
- Handles **high current & moderate voltage**
- Conducts in **forward bias**, blocks in **reverse bias**

2 Construction of Rectifier Diode (PN Junction)





Difference Between P Type and N Type Semiconductors



Materials used

- **Silicon (Si)** → most common (high efficiency)
- **Germanium (Ge)** → older, lower voltage drop

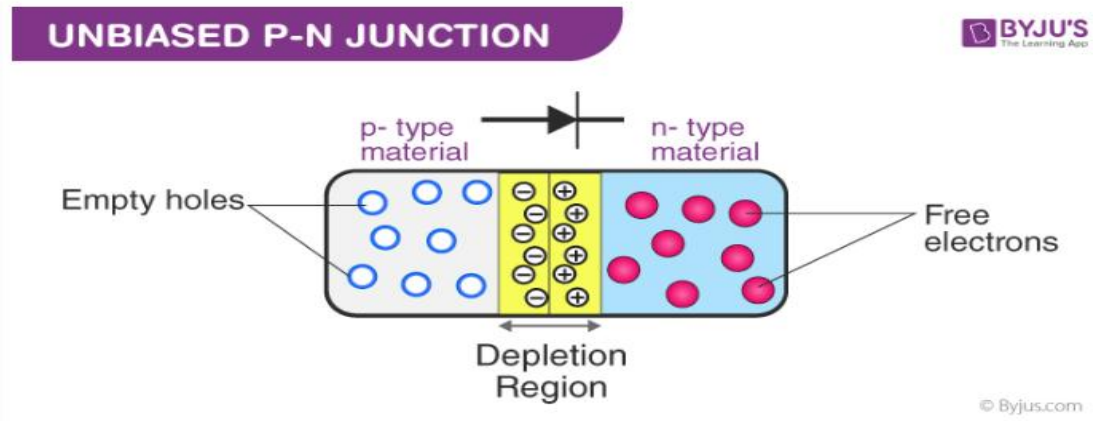
Regions

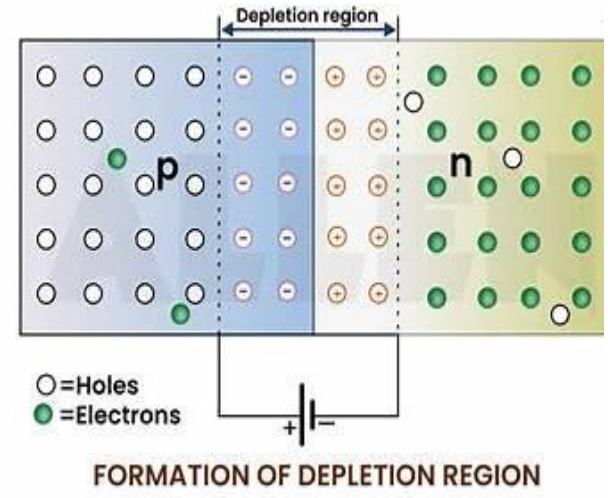
- **P-type** → holes are majority carriers
- **N-type** → electrons are majority carriers

When joined:

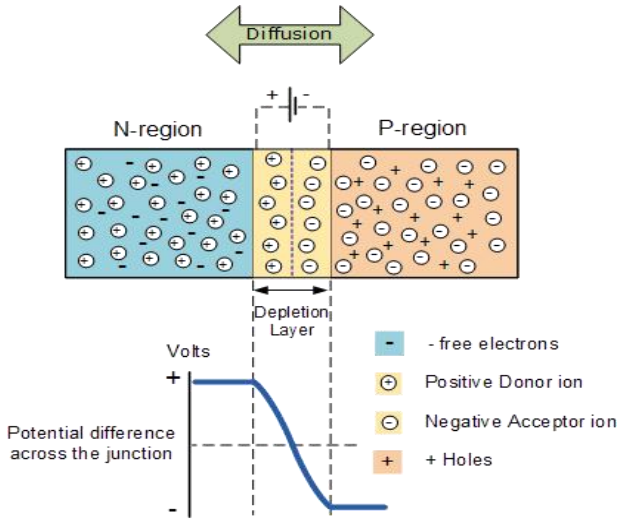
- Electrons diffuse from N → P
- Holes diffuse from P → N
- This forms the **Depletion Layer**

3 Depletion Layer (VERY IMPORTANT)





FORMATION OF DEPLETION REGION



What is it?

- A **thin region with no free charge carriers**
- Acts like an **insulating barrier**

Why it forms?

Due to **recombination of electrons and holes**

Barrier potential:

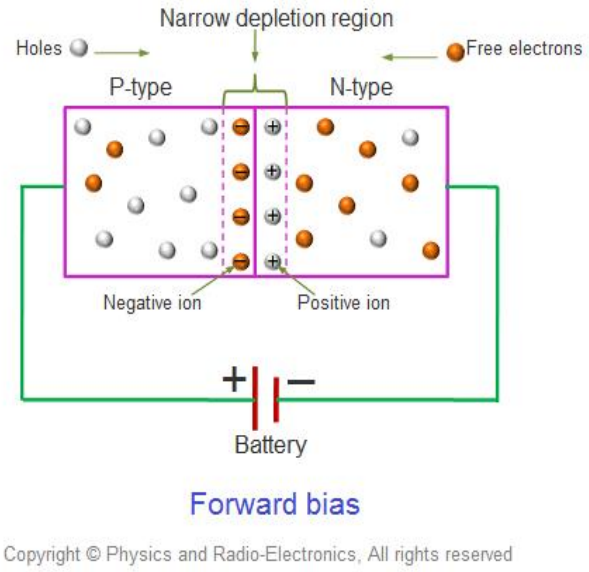
Silicon → ~0.7 V

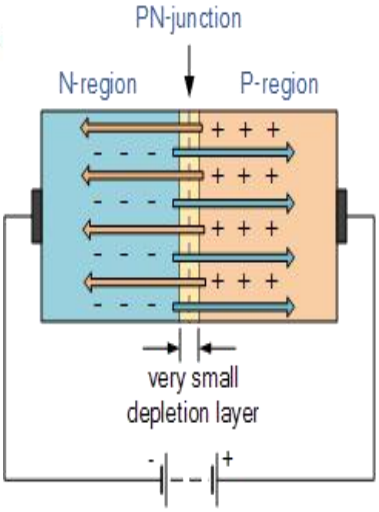
Germanium → ~0.3 V

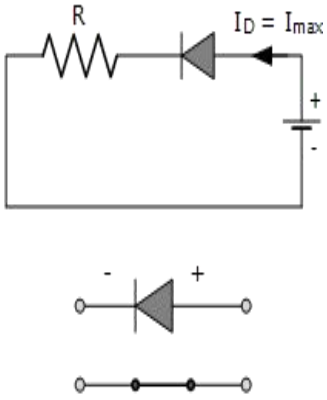
This barrier must be overcome for current to flow.

4 Biasing of Rectifier Diode (CORE WORKING LOGIC)

Forward Bias (ON State)







Connection:

P-type → Positive terminal

N-type → Negative terminal

What happens:

Depletion layer **shrinks**

Barrier potential **reduces**

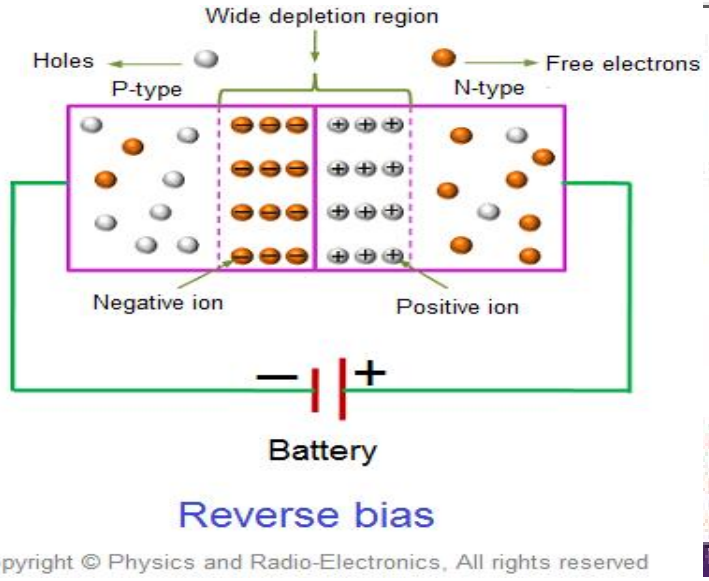
Diode **conducts**

Result:

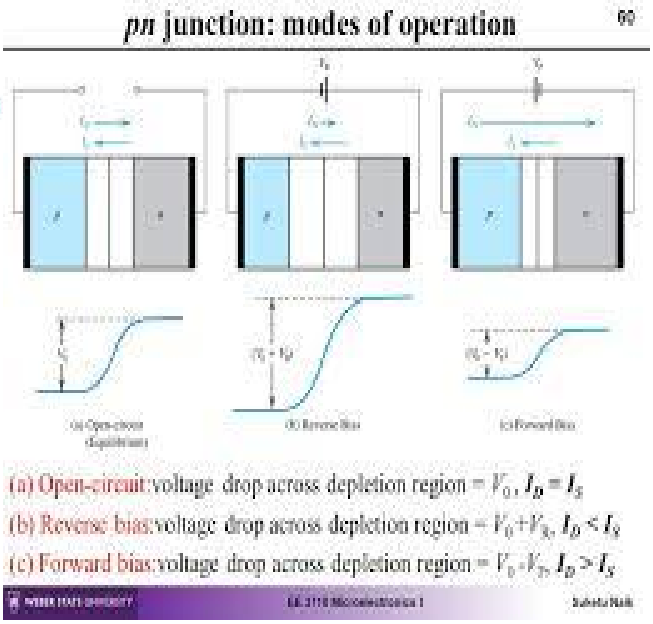
Large current flows

Diode acts like a **closed switch**

Reverse Bias (OFF State)



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Connection:

P-type → Negative terminal

N-type → Positive terminal

What happens:

Depletion layer **widens**

Barrier potential **increases**

Diode **blocks current**

Result:

Only tiny **leakage current**

Diode acts like an **open switch**

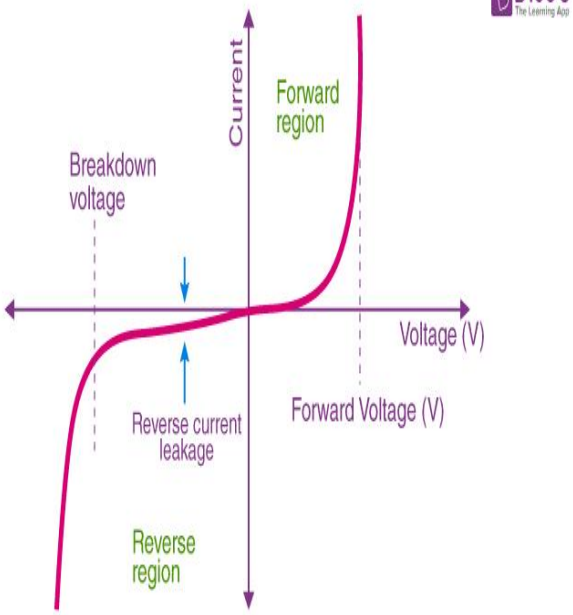
5 Operating Principle (EXAM FRIENDLY)

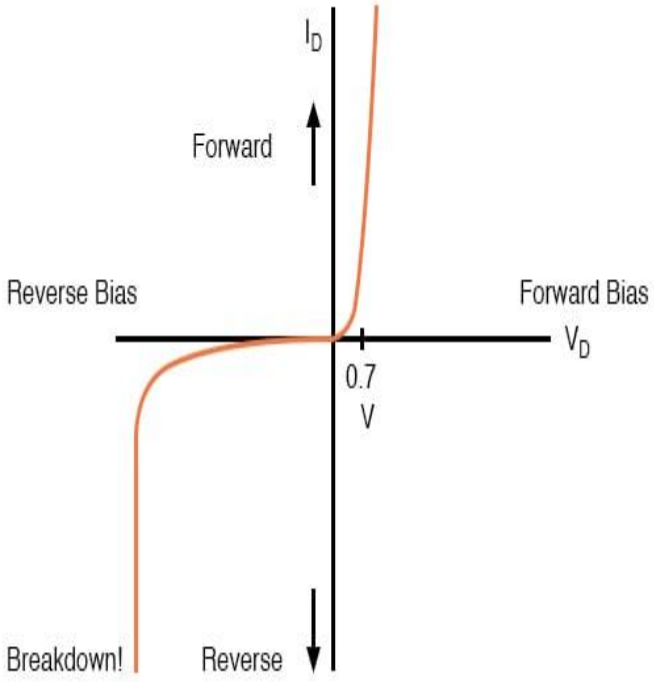
Principle:

A rectifier diode operates on the principle that **current flows easily in forward bias and is blocked in reverse bias due to the depletion layer and barrier potential of a PN junction**.

---

6 V–I Characteristics of Rectifier Diode





Forward Characteristics

No current until **cut-in voltage**

After cut-in: Current rises **exponentially**

Cut-in voltage:

Silicon → **0.7 V**

Germanium → **0.3 V**

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Reverse Characteristics

Very small **reverse saturation current**

At high reverse voltage → **breakdown**

Rectifier diodes are designed to **avoid breakdown region**

Practical Approximation (Used in Circuits)

Silicon diode ON:

- VD≈0.7VV_D \approx 0.7VVD ≈0.7V

8 Rectifier Applications (USES)

Main uses

AC → DC conversion

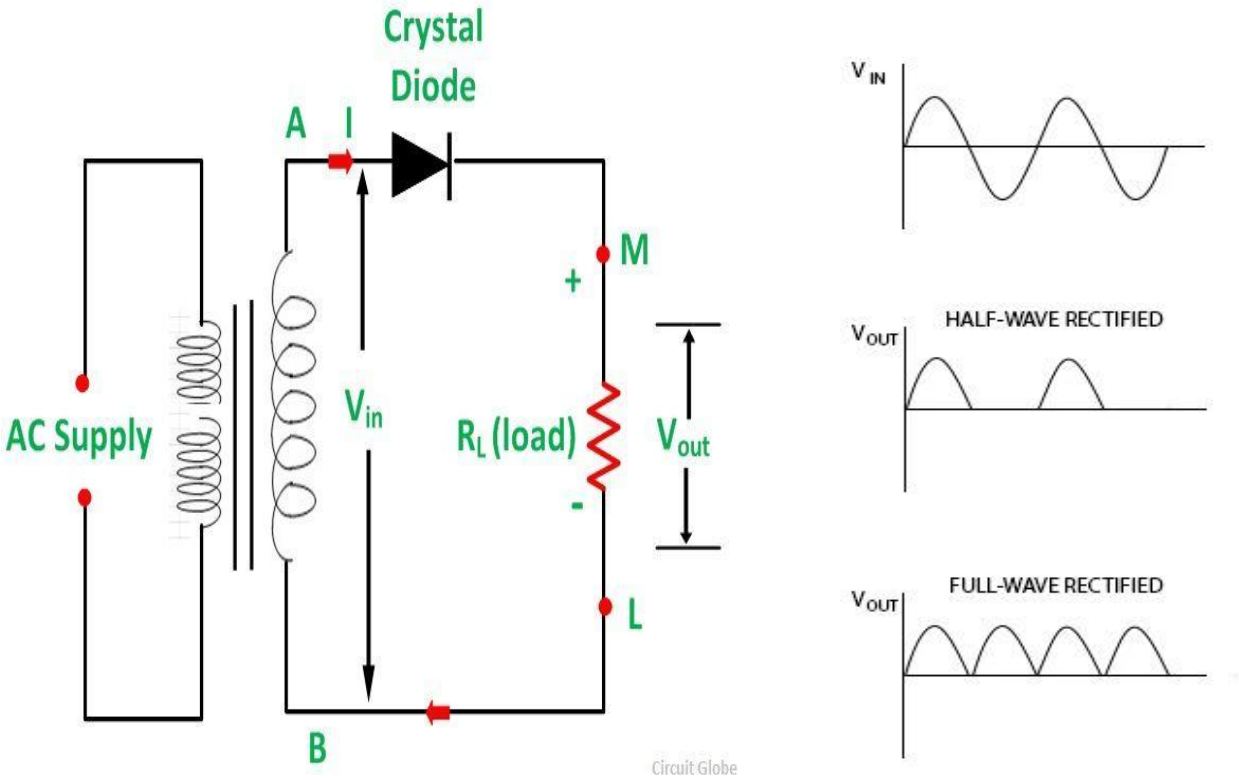
Power supplies

Battery chargers

Adapter circuits

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Example 1: Half-Wave Rectifier



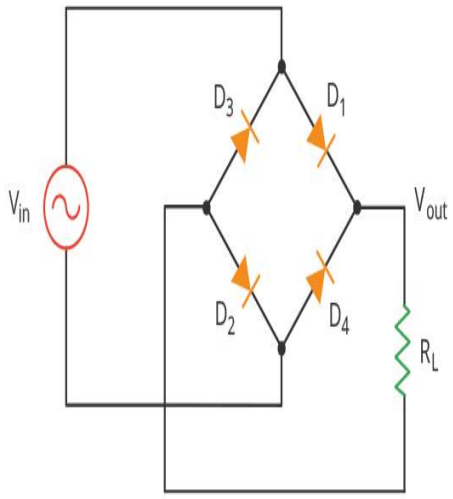
Working:

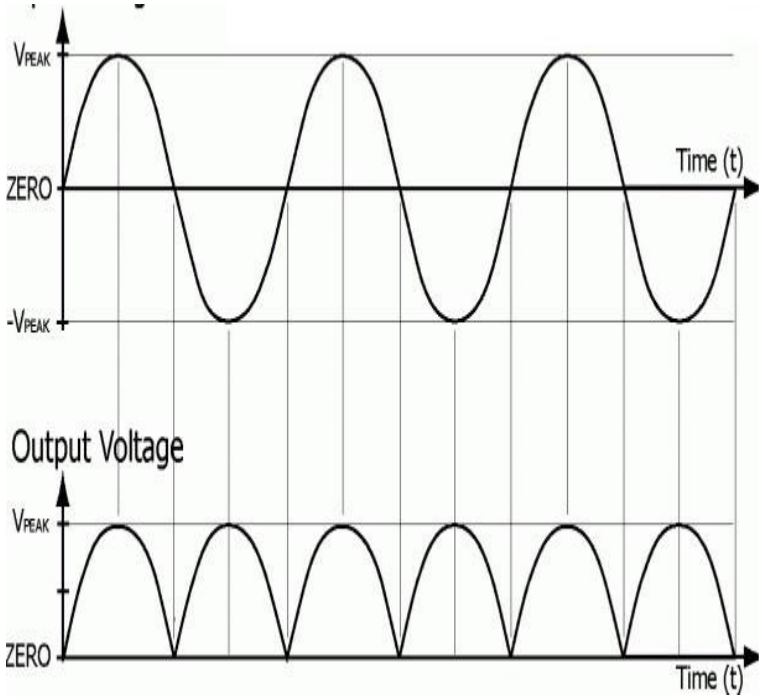
Positive half cycle → diode conducts

Negative half cycle → diode blocks

Output: Pulsating DC

Example 2: Full-Wave Rectifier





Working:

Both half cycles are used

Output frequency = **2 × input frequency**

Output: Smoother DC

---

9 Output Waveforms (VERY IMPORTANT)

Half-Wave Rectifier

Output only during **positive cycles**

Low efficiency

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Full-Wave Rectifier

Output during **both cycles**

Higher efficiency

Less ripple

Advantages & Limitations

Advantages

Simple construction

Low cost

Reliable

High current handling

Limitations

Power loss due to 0.7V drop

Generates ripple

Not suitable for high-frequency switching

Key Points

Rectifier diode is a PN junction device

Conducts in forward bias, blocks in reverse bias

Depletion layer controls current flow

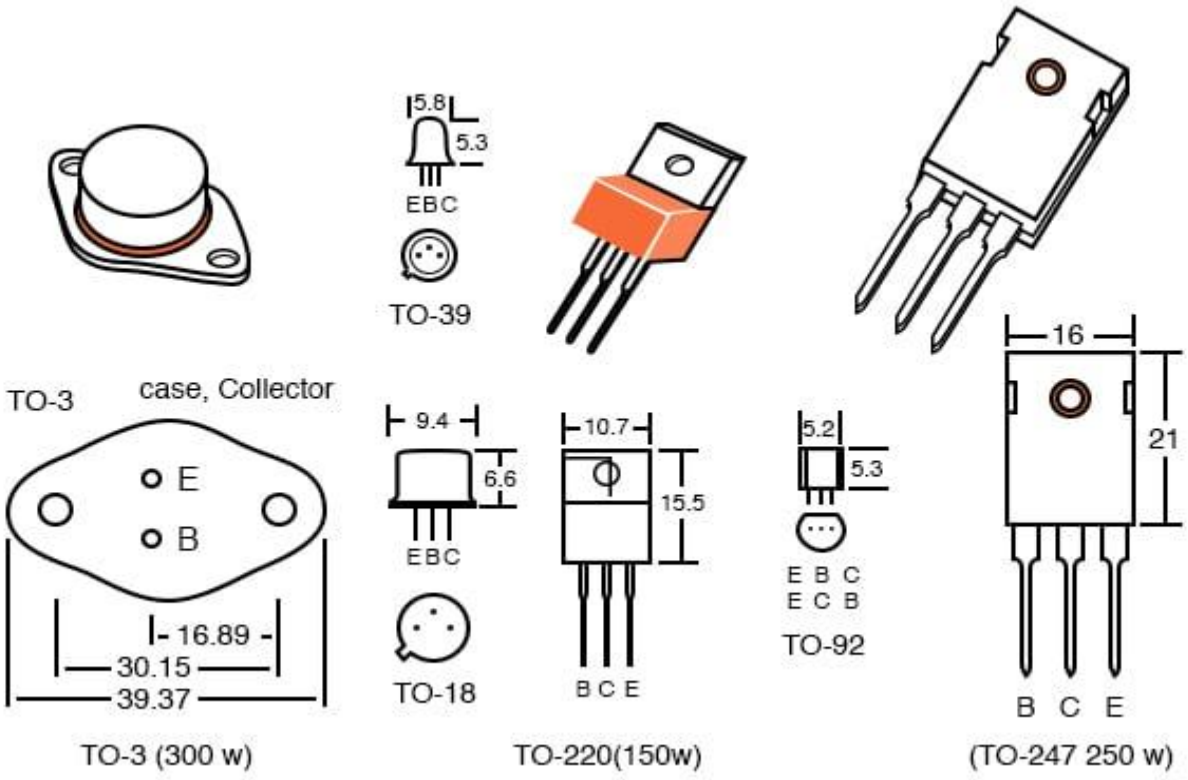
Used for AC to DC conversion

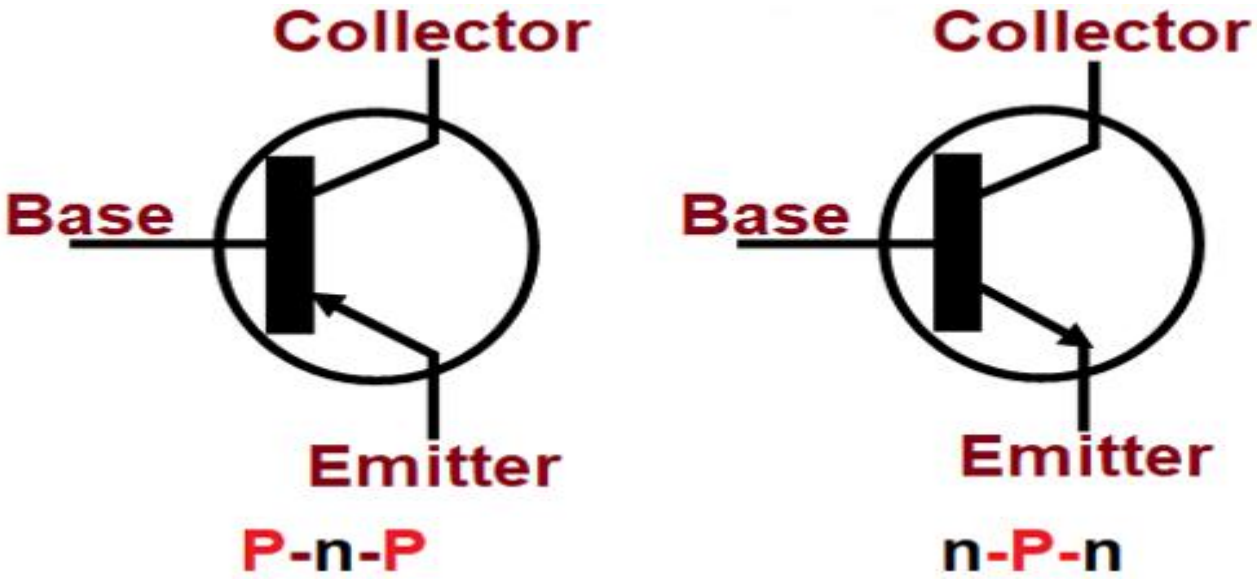
Silicon diodes have 0.7V drop

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Bipolar Junction Transistor (BJT – Power BJT)

1 What is a Power BJT?





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Definition

A **Power Bipolar Junction Transistor (BJT)** is a **current-controlled semiconductor device**

Used mainly as a **high-power electronic switch**

Called **bipolar** because **both electrons and holes participate in conduction**

Why "power" BJT?

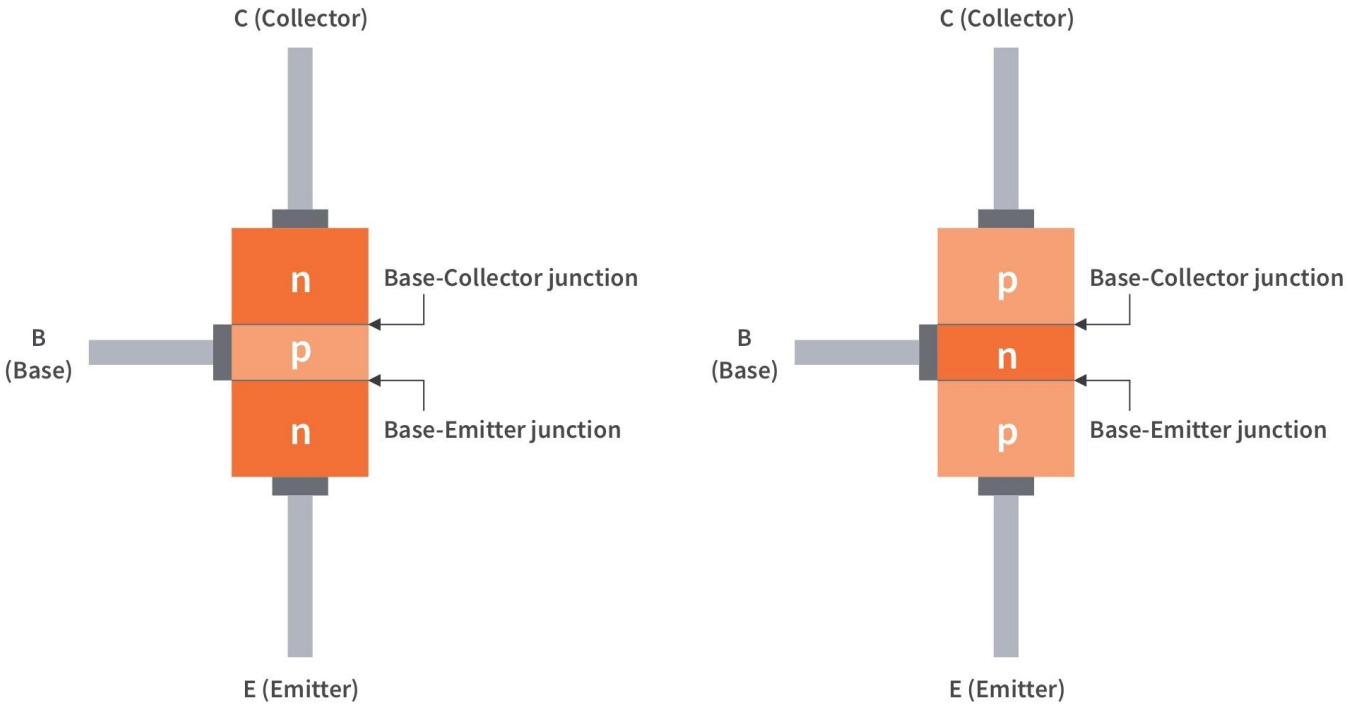
Handles **high current**

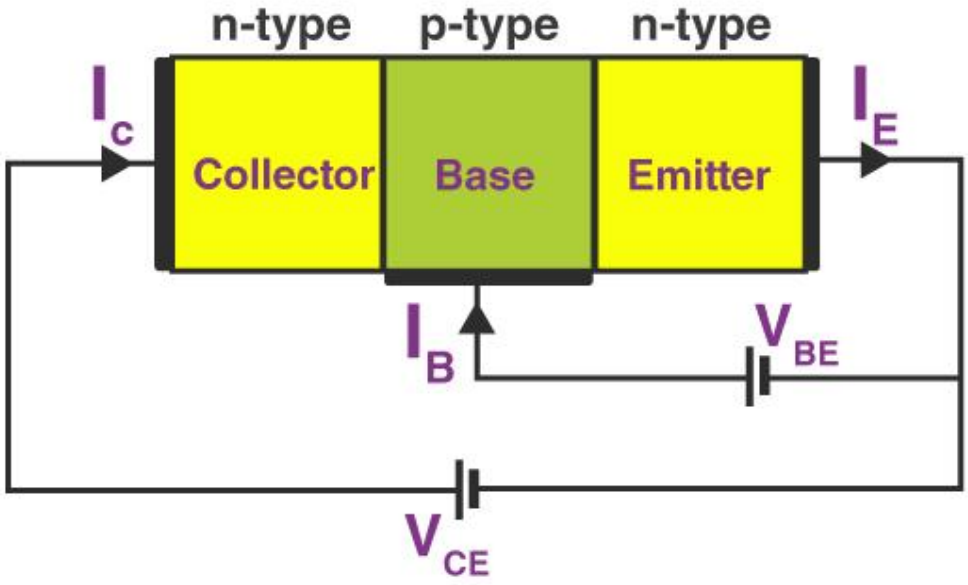
Handles **moderate to high voltage**

Used in **motor drives, inverters, power amplifiers**

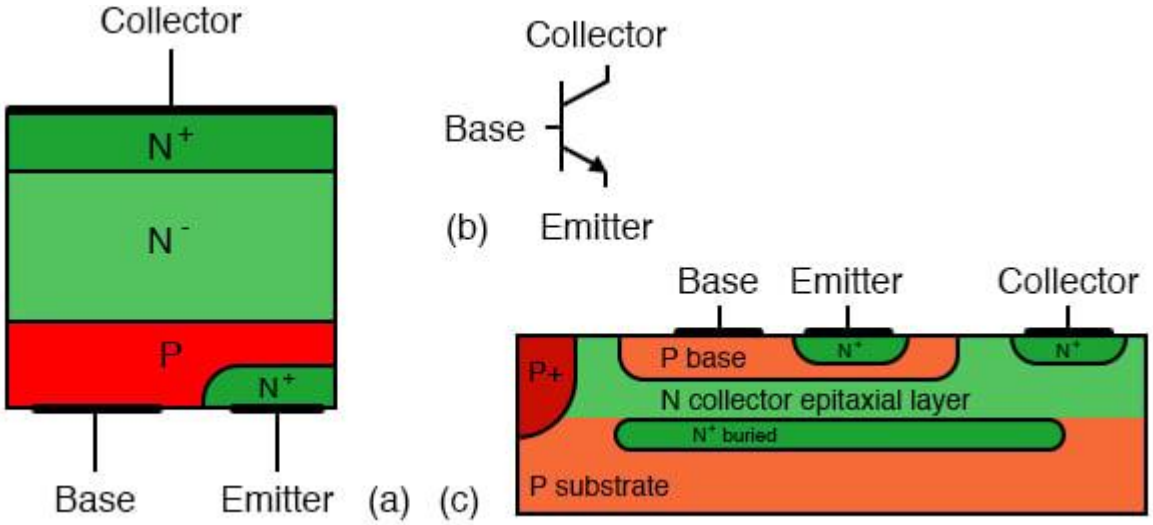
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2 Construction of Power BJT (VERY IMPORTANT





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Terminals

Emitter (E) – heavily doped

Base (B) – very thin, lightly doped

Collector (C) – moderately doped, large area

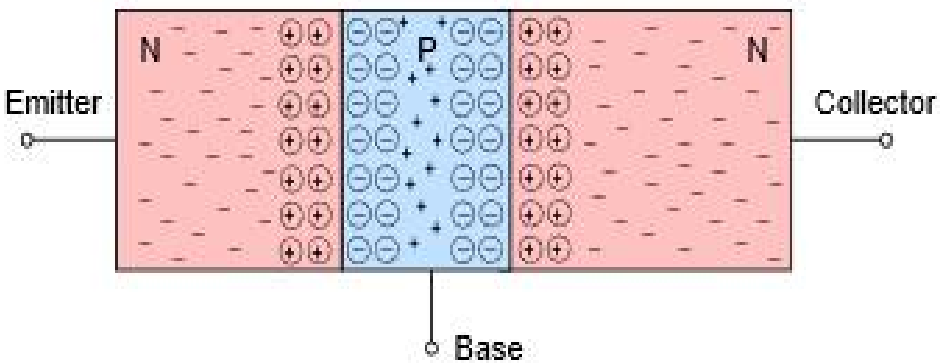
PN Junctions inside BJT

Emitter–Base (EB) junction

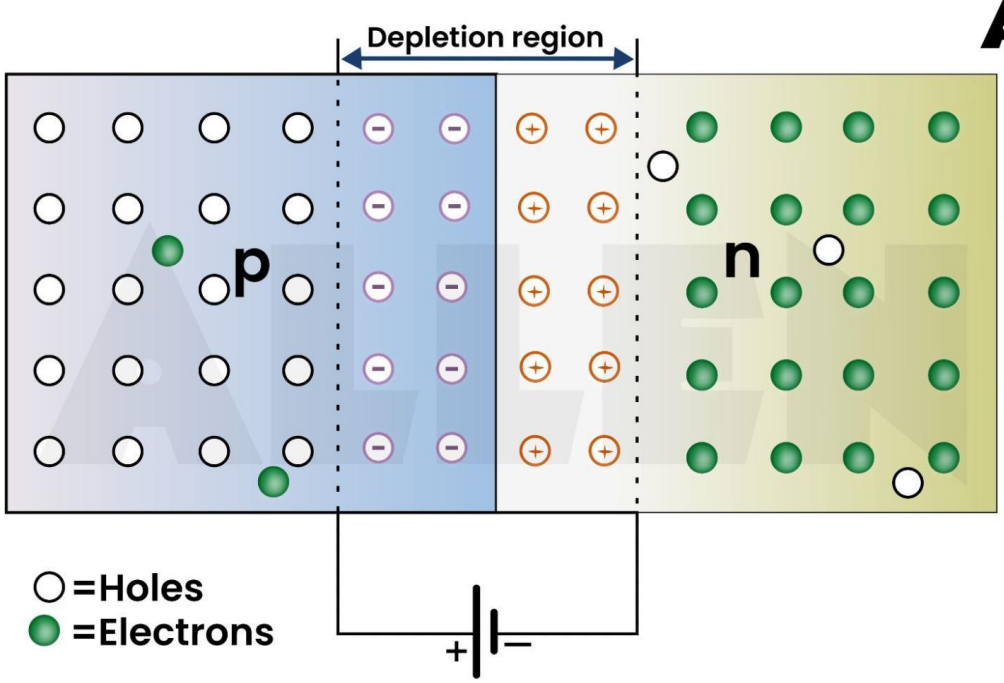
Base–Collector (BC) junction

So a BJT has **TWO PN junctions**

3 Depletion Layer in Power BJT



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FORMATION OF DEPLETION REGION

What is depletion layer?

A region **without free charge carriers**

Formed due to **recombination of electrons and holes**

In BJT

Depletion layer exists at:

EB junction

BC junction

Role

Controls **current flow**

Width changes with **biasing**

Key factor in **switching speed**

4 Operating Principle of Power BJT (CORE IDEA)

A small base current controls a much larger collector current

Why this happens?

EB junction is **forward biased**

BC junction is **reverse biased**

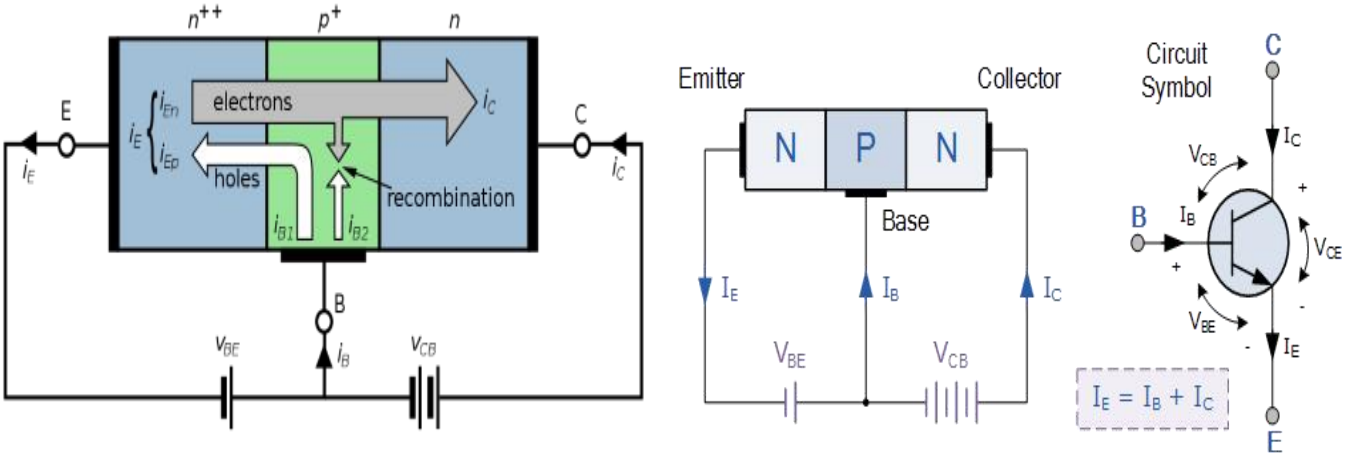
Charge carriers injected from emitter are **pulled into collector**

This makes BJT:

A **current amplifier**

A **controlled power switch**

5 Working of Power BJT (STEP-BY-STEP)



Step-1: Forward bias EB junction

Base slightly positive w.r.t emitter

Electrons injected from emitter into base

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Step-2: Thin base region

Very little recombination

Most electrons reach collector junction

Step-3: Reverse biased BC junction

Collector attracts electrons

Large collector current flows

✔ **Collector current depends on base current**

---

6 Modes of Operation (SWITCHING VIEW)

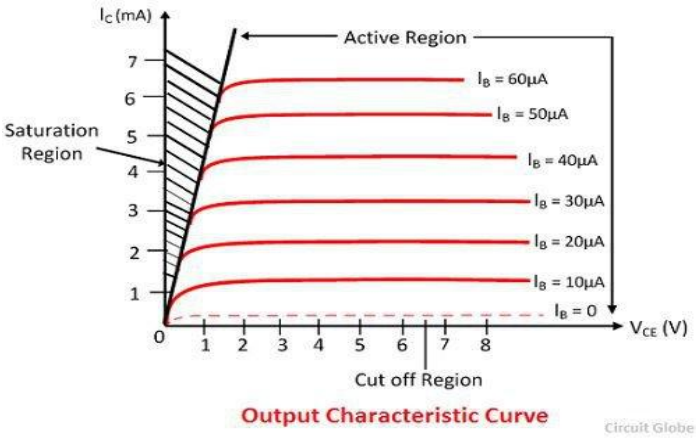
Mode	EB Junction	BC Junction	State
Cut-off	Reverse	Reverse	OFF
Active	Forward	Reverse	Amplification
Saturation	Forward	Forward	ON

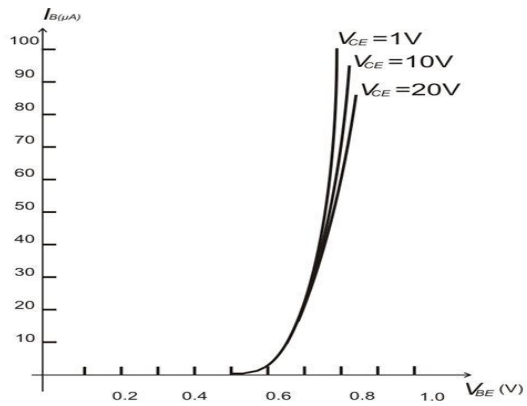
Power BJTs are used ONLY in:

Cut-off (OFF)

Saturation (ON)

7 V–I Characteristics of Power BJT





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Input Characteristics

Plot: **IB vs VBE**

Similar to diode curve

VBE ≈ **0.7 V (Silicon)**

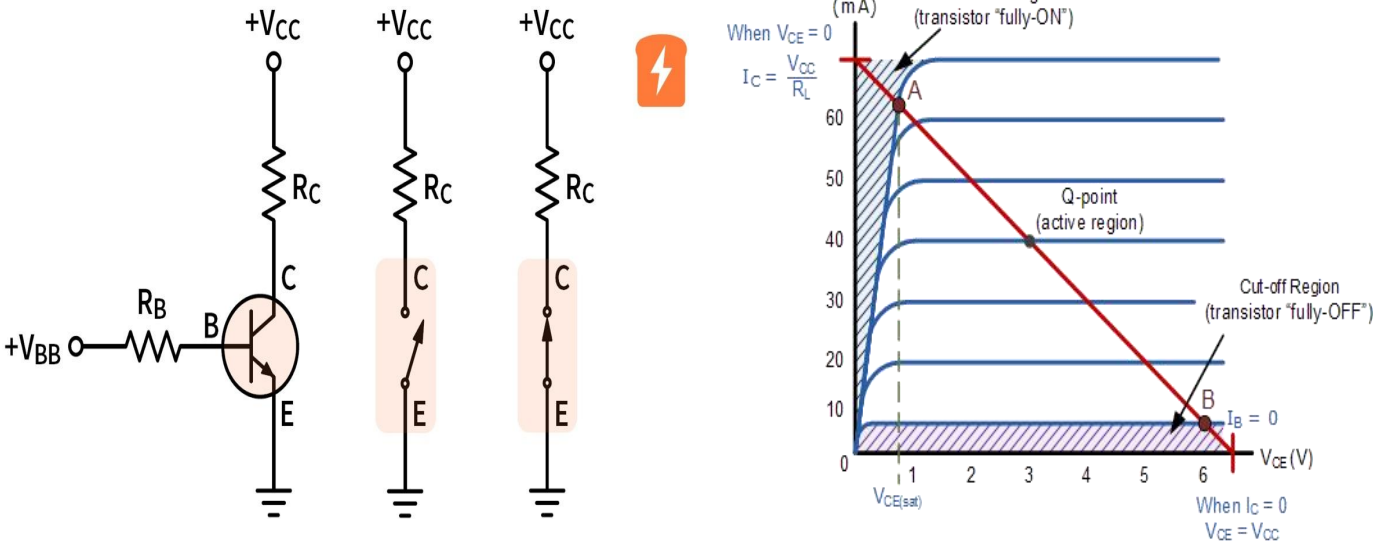
Output Characteristics

Plot: **IC vs VCE**

Saturation region → **switch ON**

Cut-off region → **switch OFF**

8 Power BJT as a Switch (MOST IMPORTANT)



OFF State (Cut-off)

IB = 0

Both junctions reverse biased

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IC ≈ 0

VCE = VCC

Acts like **open switch**

ON State (Saturation)

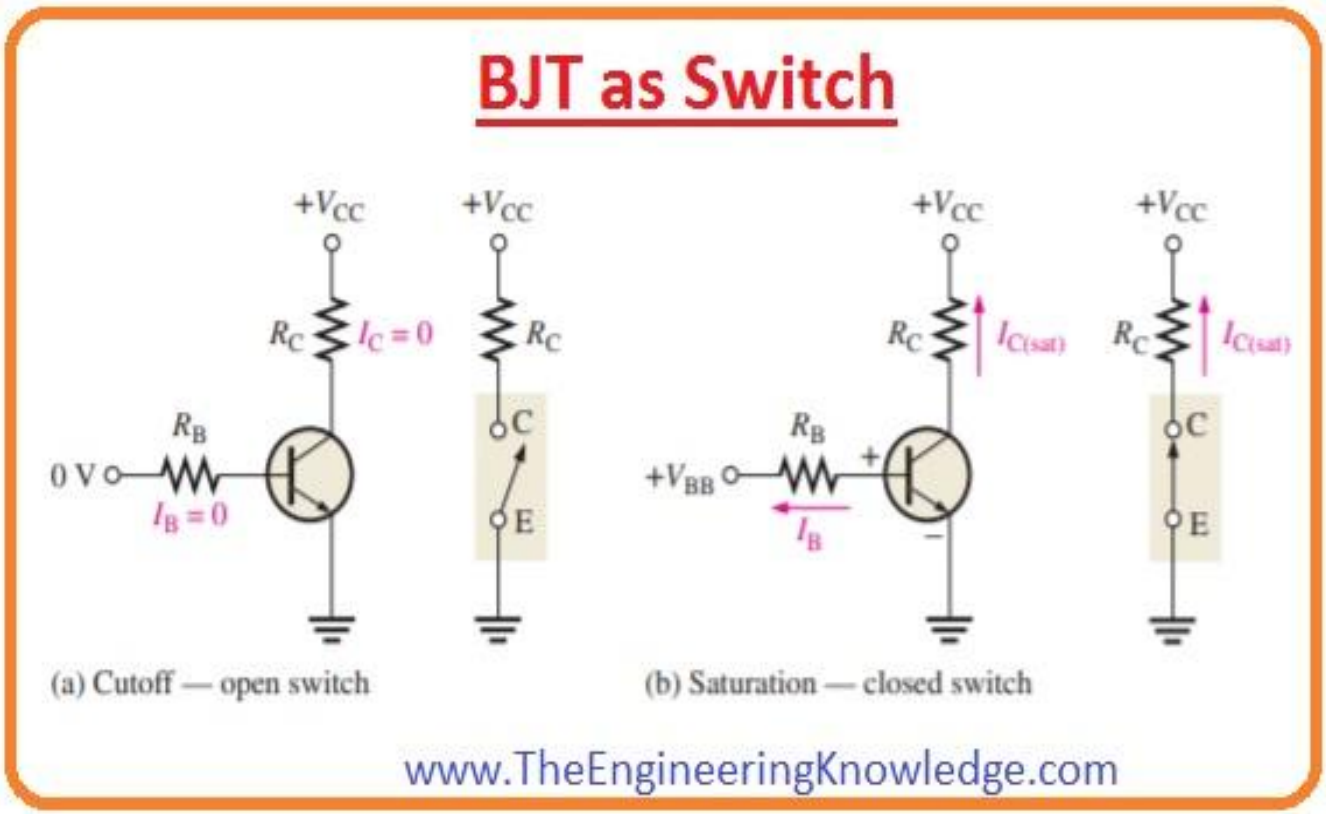
Sufficient base current applied

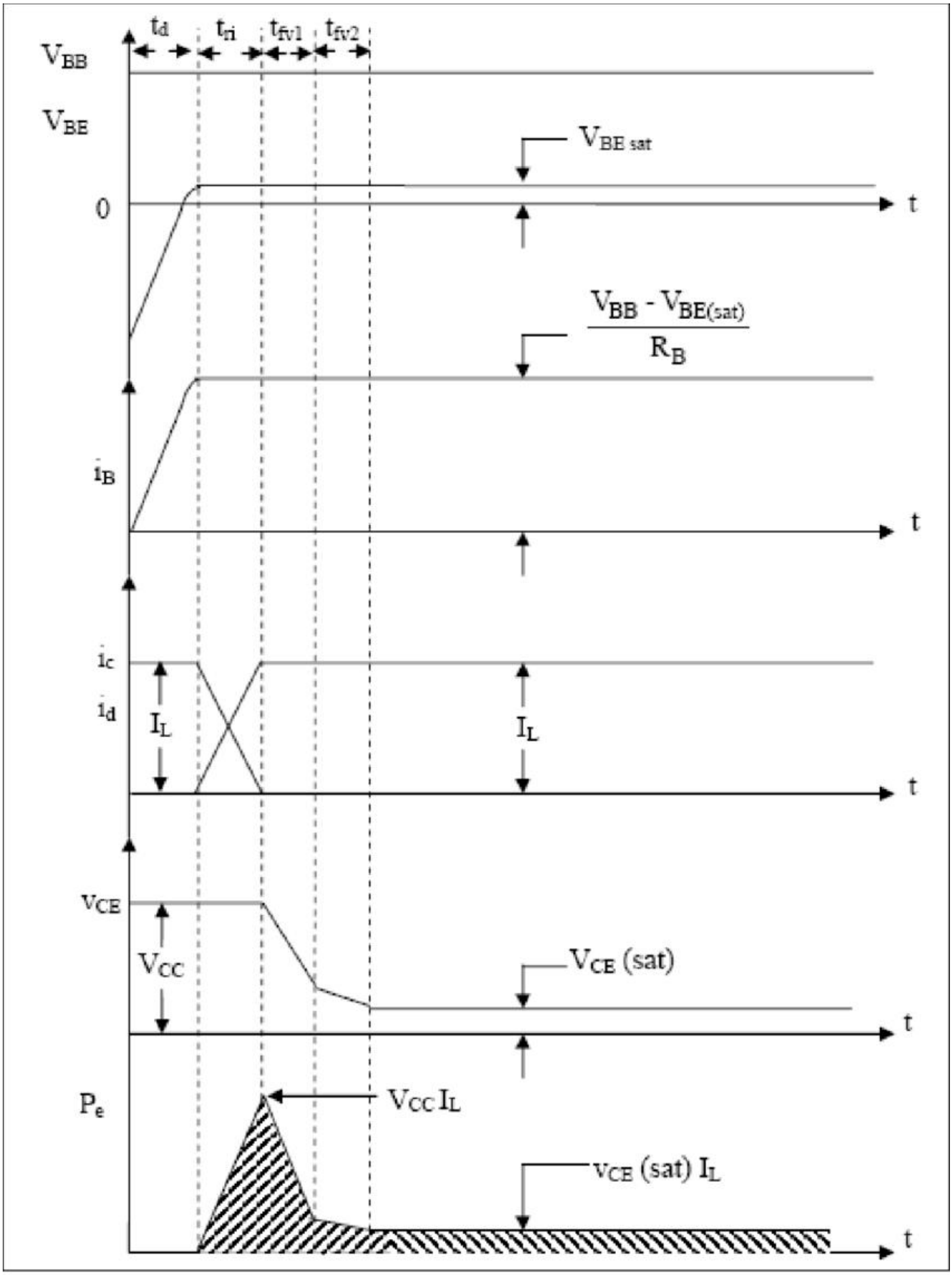
Both junctions forward biased

VCE(sat) ≈ **0.2 V**

Acts like **closed switch**

9 Switching Operation (STEP-BY-STEP)





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Turn-ON Process

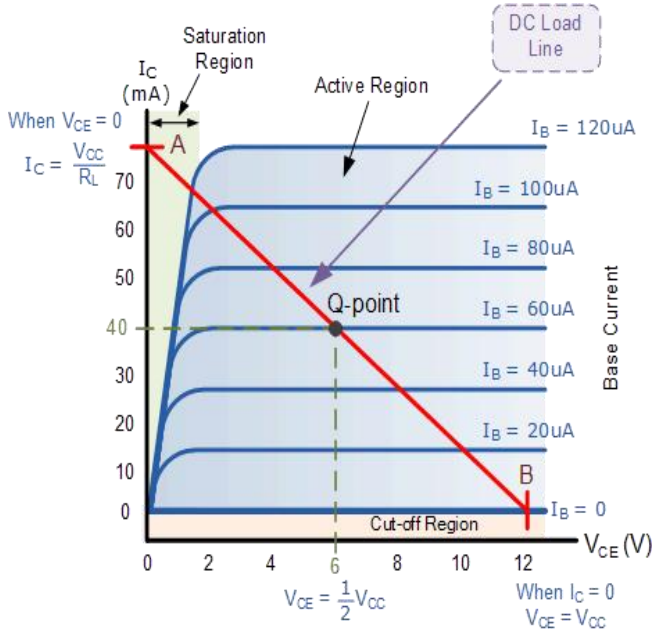
1. Base current applied
2. EB depletion layer reduces
3. BC junction forward biased
4. Transistor enters saturation
5. Load current flows

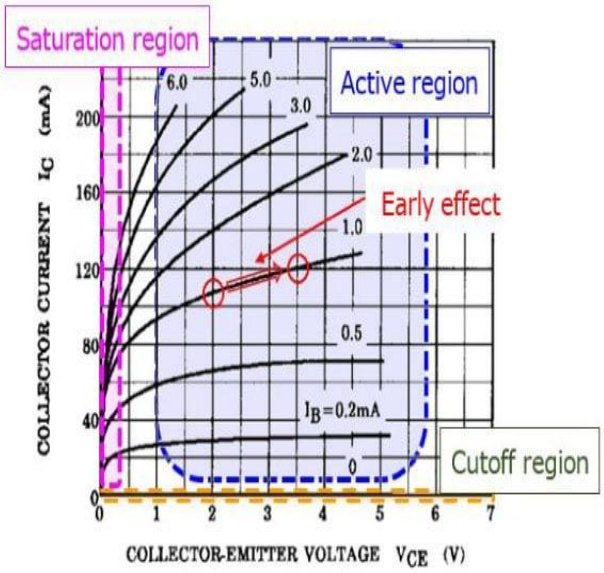
Turn-OFF Process

1. 1.Base current removed
2. Stored charge remains
3. Recombination occurs
4. Depletion layers widen
5. Current stops (slow OFF)

Stored charge causes slow turn-OFF (major drawback)

Switching Waveforms (EXAM FAVORITE)





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Waveforms explained

IB → control signal

IC → follows IB

VCE

High during OFF

Drops during ON

Key times:

1. Delay time
2. Rise time
3. Fall time

1 3 Uses of Power BJT (2 EXAMPLES)

Example 1: DC Motor Control

BJT acts as ON/OFF switch

Controls motor speed

Example 2: Inverter Circuits

Switches DC at high current

Produces AC output

1 4 Switch Basics

What is a switch?

A device with **only two state**

ON → conducts

OFF → blocks

How BJT works as a switch

Base current = **control**

Collector current = **power path**

Fully controlled device (ON & OFF controlled)

1 5 Advantages & Limitations

✅ Advantages

Simple structure

High current capability

Good for linear amplification

❌ Limitations

Needs continuous base current

Slow switching

High switching losses

FINAL SUMMARY

Power BJT is a **current-controlled device**

Has **two PN junctions**

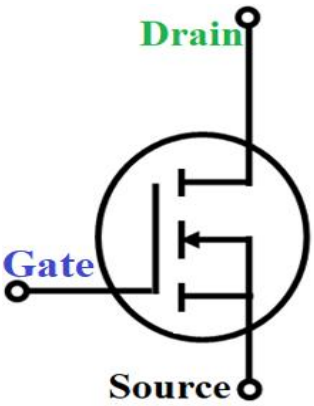
Depletion layer controls conduction

Used mainly in **cut-off & saturation**

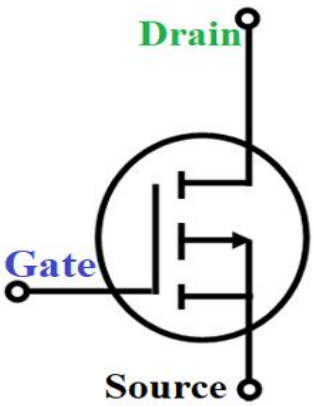
Acts as a **fully controlled power switch**

POWER MOSFET (Metal–Oxide–Semiconductor Field-Effect Transistor)

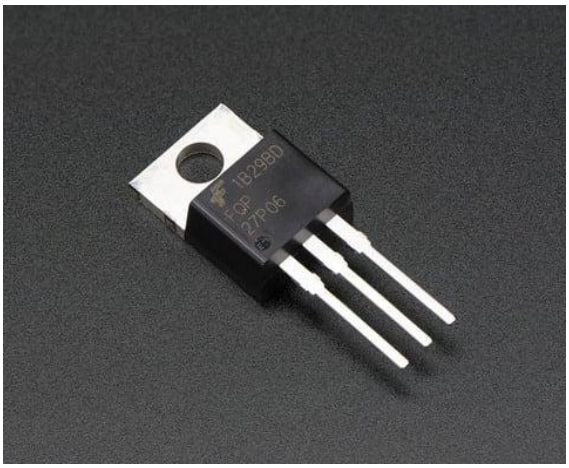
1 What is a Power MOSFET?

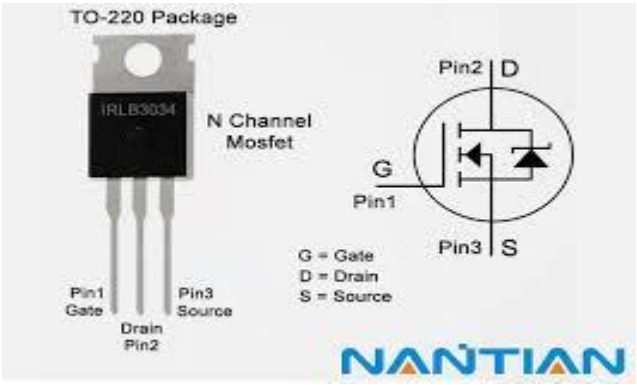


N-Channel MOSFET



P-Channel MOSFET





Definition

A **Power MOSFET** is a **voltage-controlled, unipolar semiconductor device**

Used mainly as a **high-speed power electronic switch**

Conduction occurs due to **only one type of charge carrier** (electrons or holes)

Why MOSFET is preferred over BJT

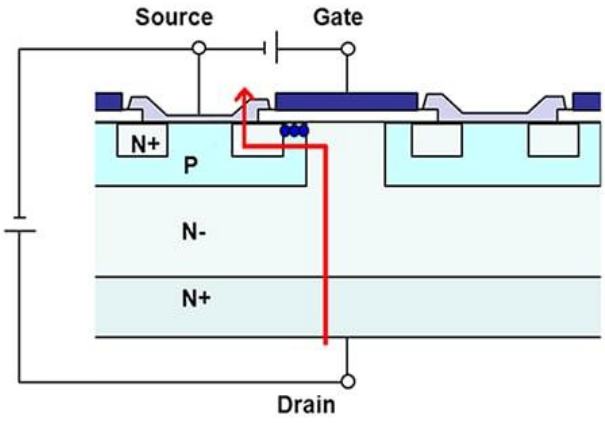
No continuous control current

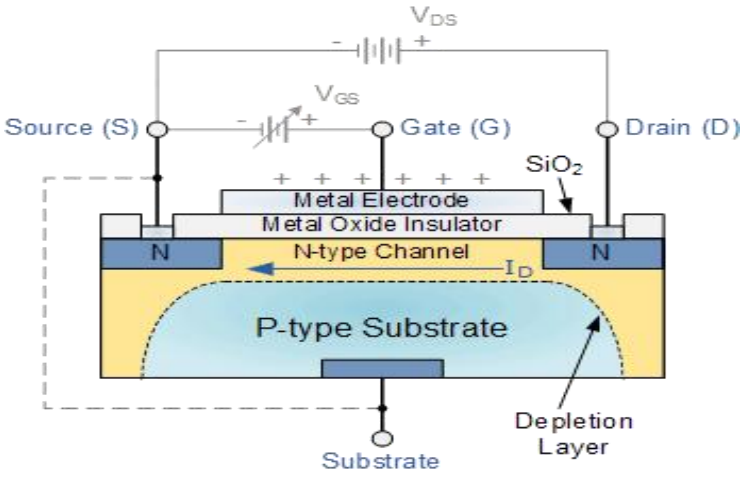
Very fast switching

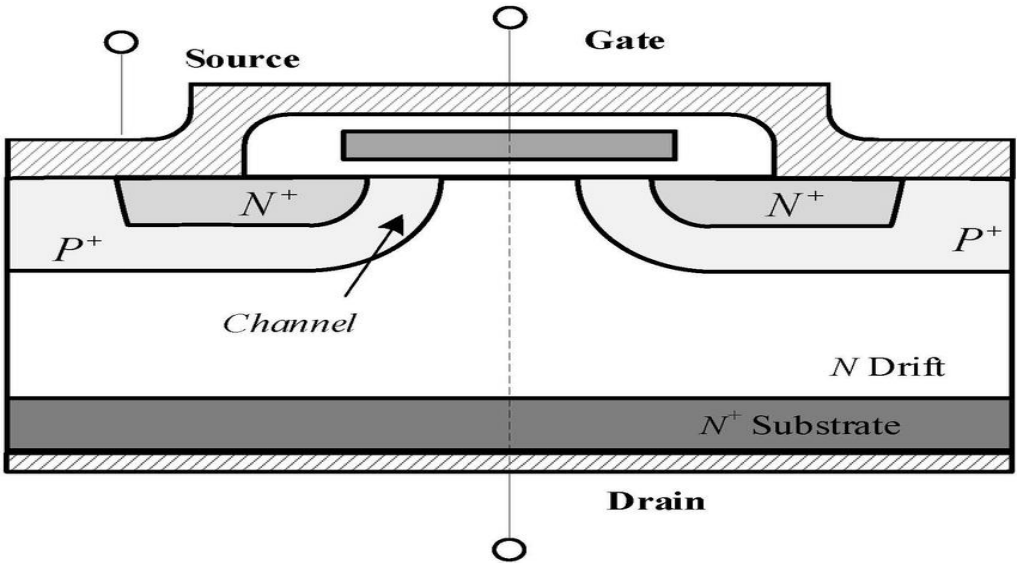
High efficiency

Easy gate drive

2 Construction of Power MOSFET (CORE FOUNDATION)







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Terminals

Gate (G) – control terminal

Drain (D) – output terminal

Source (S) – reference terminal

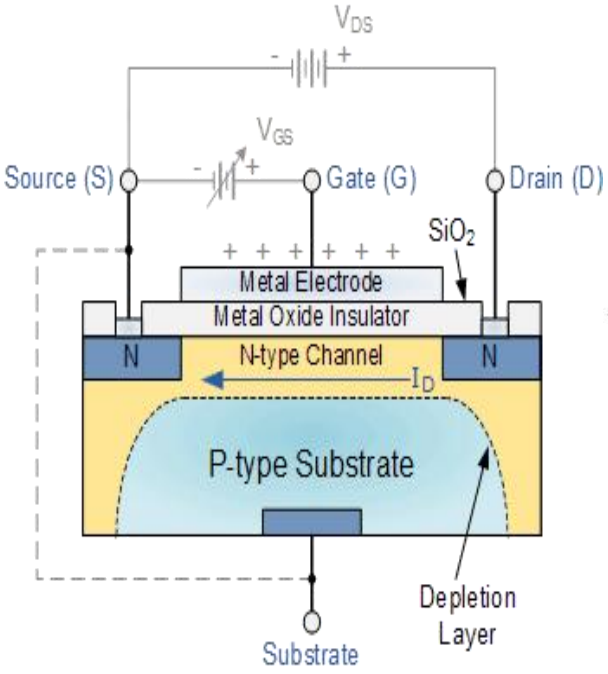
Internal structure (Power MOSFET)

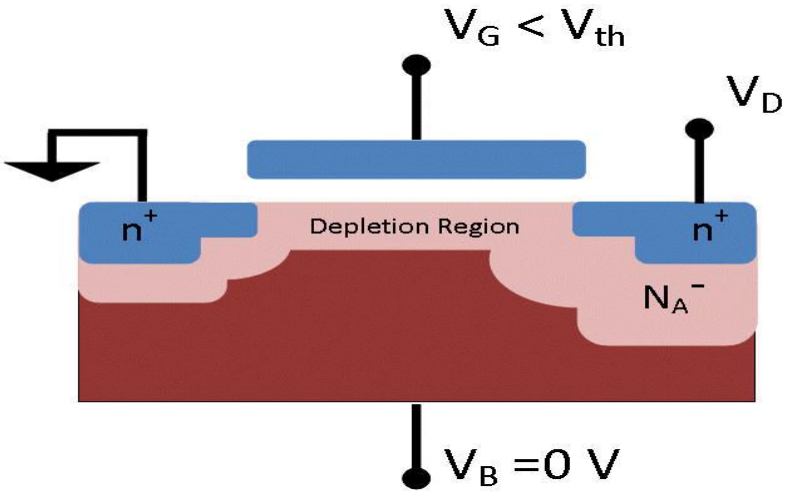
Vertical structure (for high current)

Gate insulated by **SiO_2 (oxide layer)**

Body diode inherently present

3 Junctions & Depletion Layer in MOSFET





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Important junctions

PN junction between:

Body (P-type) and Drain (N-type)

This junction creates the **body diode**

Depletion layer

Exists between:

P-body and N-drain

Controls **blocking voltage**

Width increases with **reverse bias**

Gate is insulated → no depletion layer at gate like BJT

4 Operating Principle of Power MOSFET (KEY IDEA)

An electric field created by gate voltage controls the conductivity of a channel between drain and source

Key concept

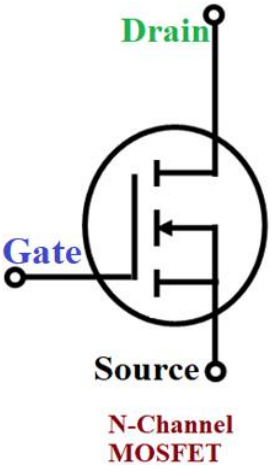
Gate voltage controls **electric field**

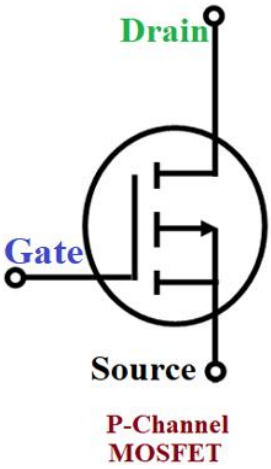
Electric field controls **carrier movement**

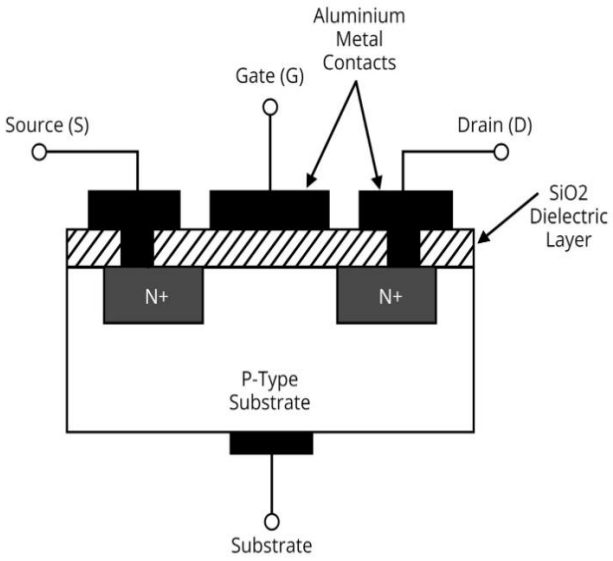
No gate current → only gate voltage

5 Types of MOSFET (VERY IMPORTANT – EXAM + INTERVIEW)

1. N-Channel MOSFET







Features

Electrons are charge carriers

Higher speed

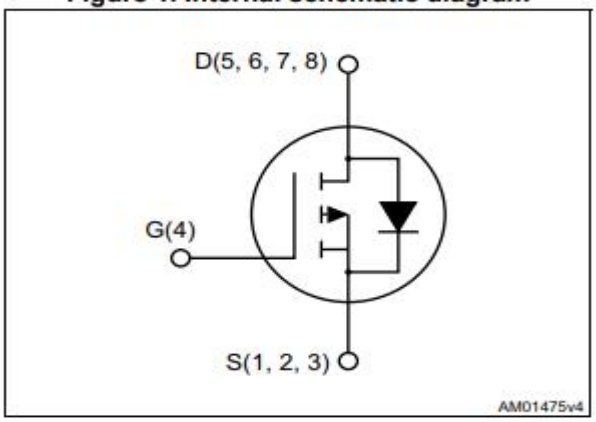
Lower RDS(on)

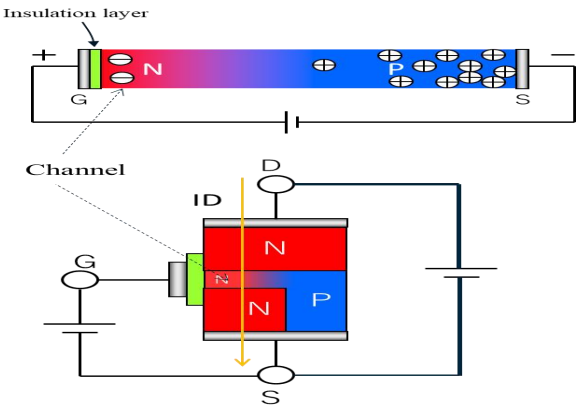
Most commonly used

2. P-Channel MOSFET

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Figure 1. Internal schematic diagram





Features

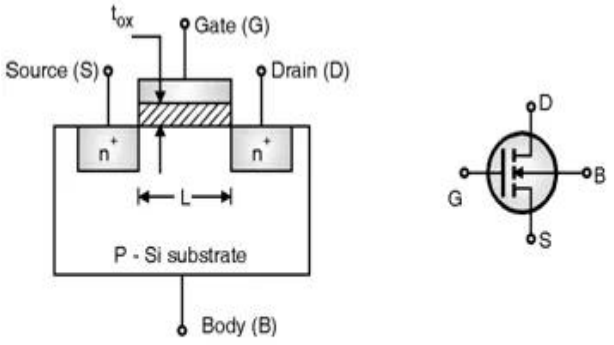
Holes are charge carriers

Slower than N-channel

Higher RDS(on)

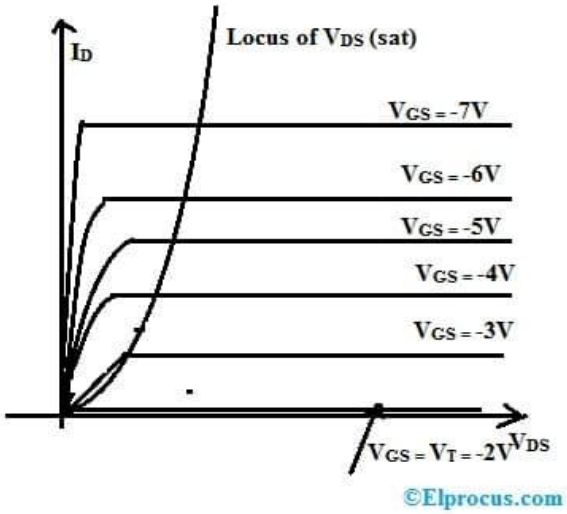
Used in high-side switching

3. Enhancement-Mode MOSFET (MOST USED)



(a) Cross sectional view (b) Symbol

n-channel enhancement mode MOSFET



Normally **OFF**

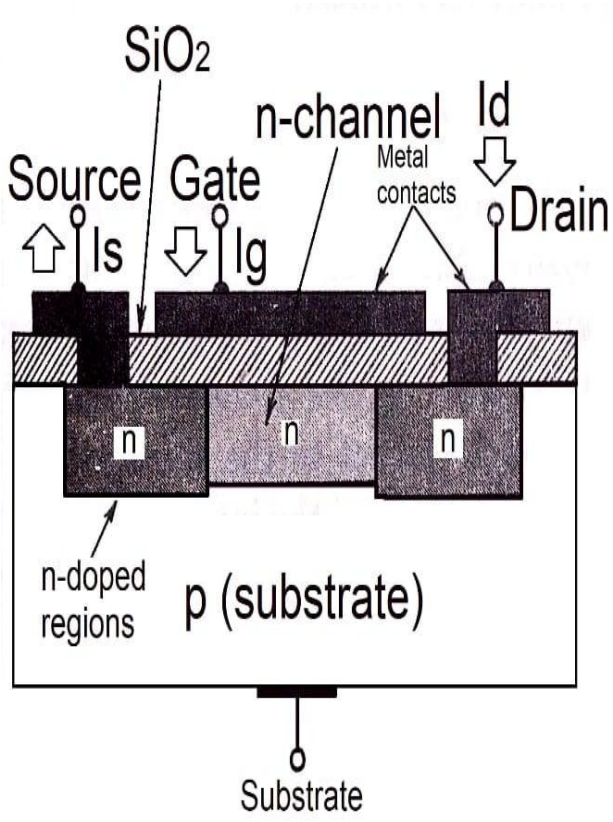
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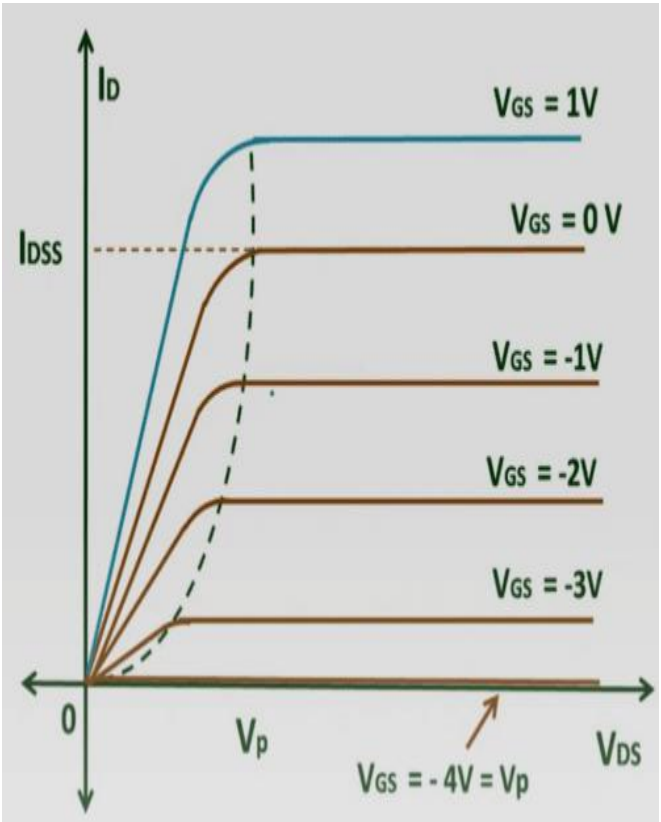
Channel forms only when:

VGS > Vth

Used in **power electronics**

4. Depletion-Mode MOSFET





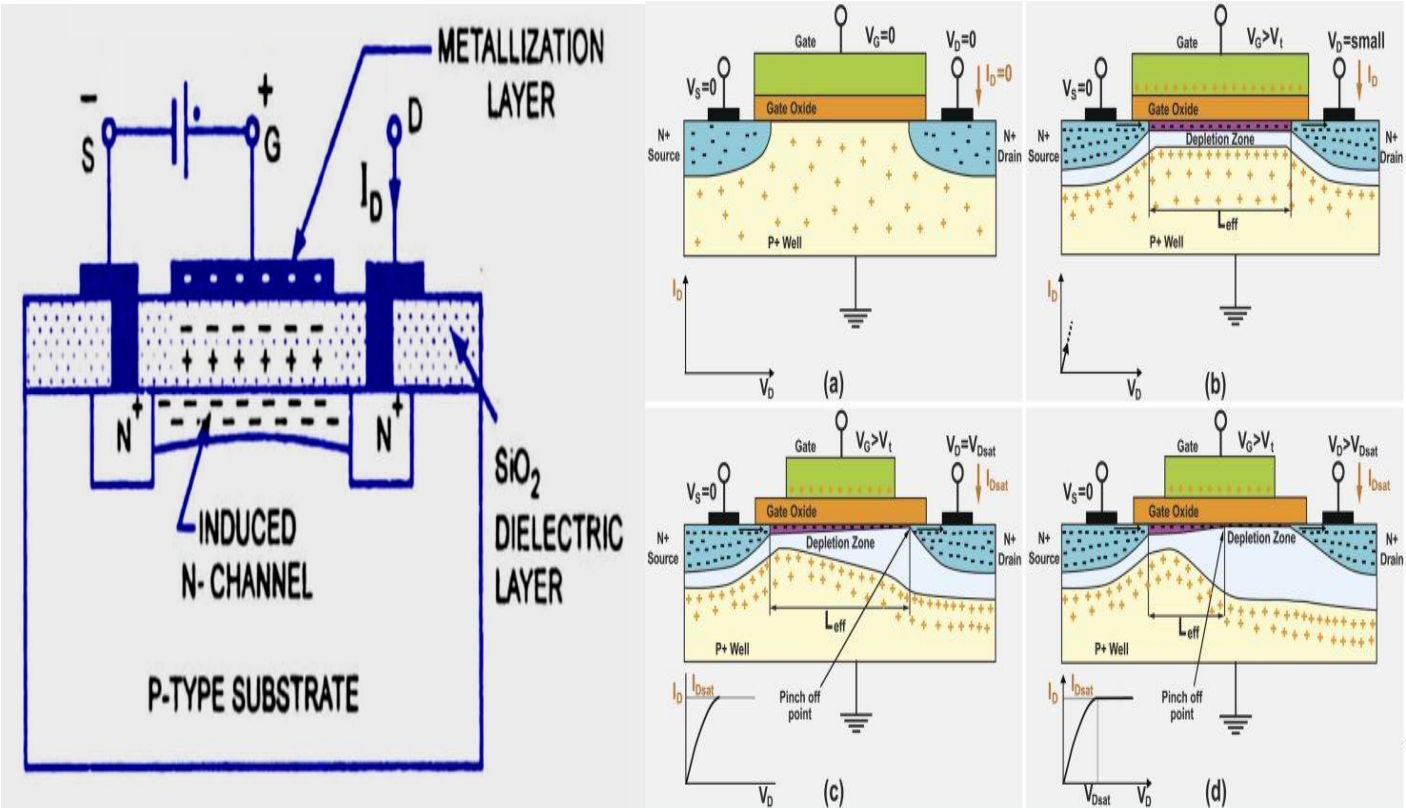
Normally **ON**

Gate voltage reduces channel

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Rare in power applications

6 Working of N-Channel Enhancement Power MOSFET

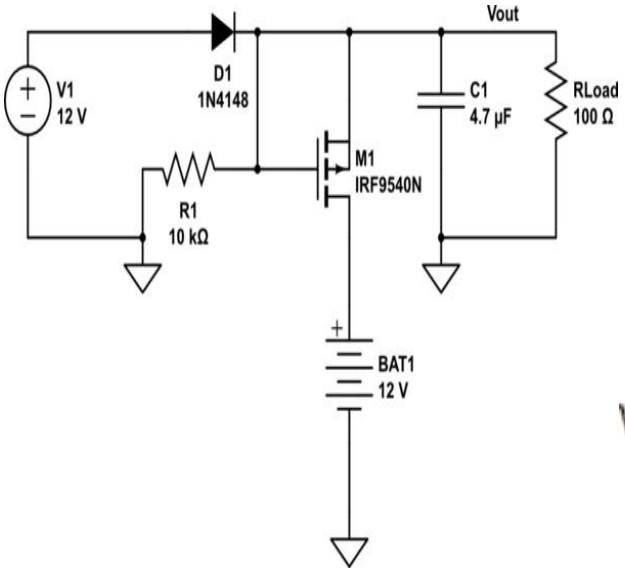


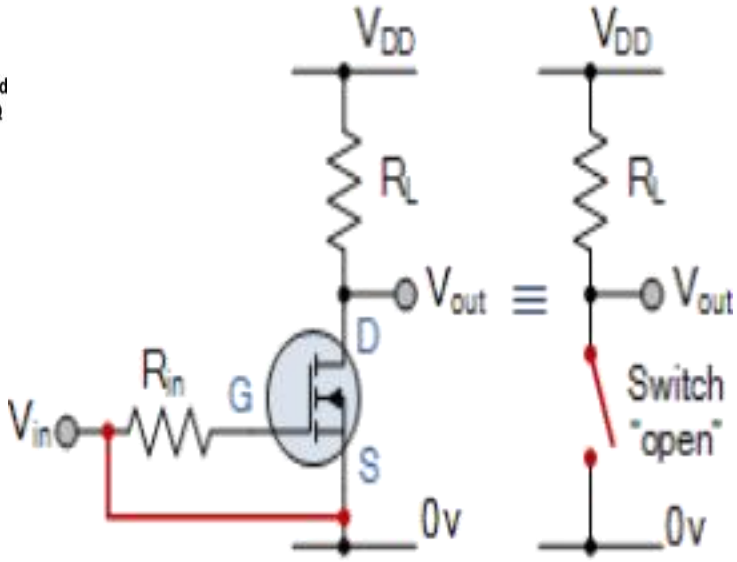
Step-by-step working

1. VGS = 0 → no channel → OFF
2. VGS > Vth → electric field forms
3. Electrons attracted to gate region
4. Conductive **N-channel forms**
5. Drain current flows

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7 MOSFET as a Switch (MOST IMPORTANT)





OFF State

VGS < Vth

No channel

IDS ≈ 0

Acts like **open switch**

ON State

VGS ≥ 10–12 V (power MOSFET)

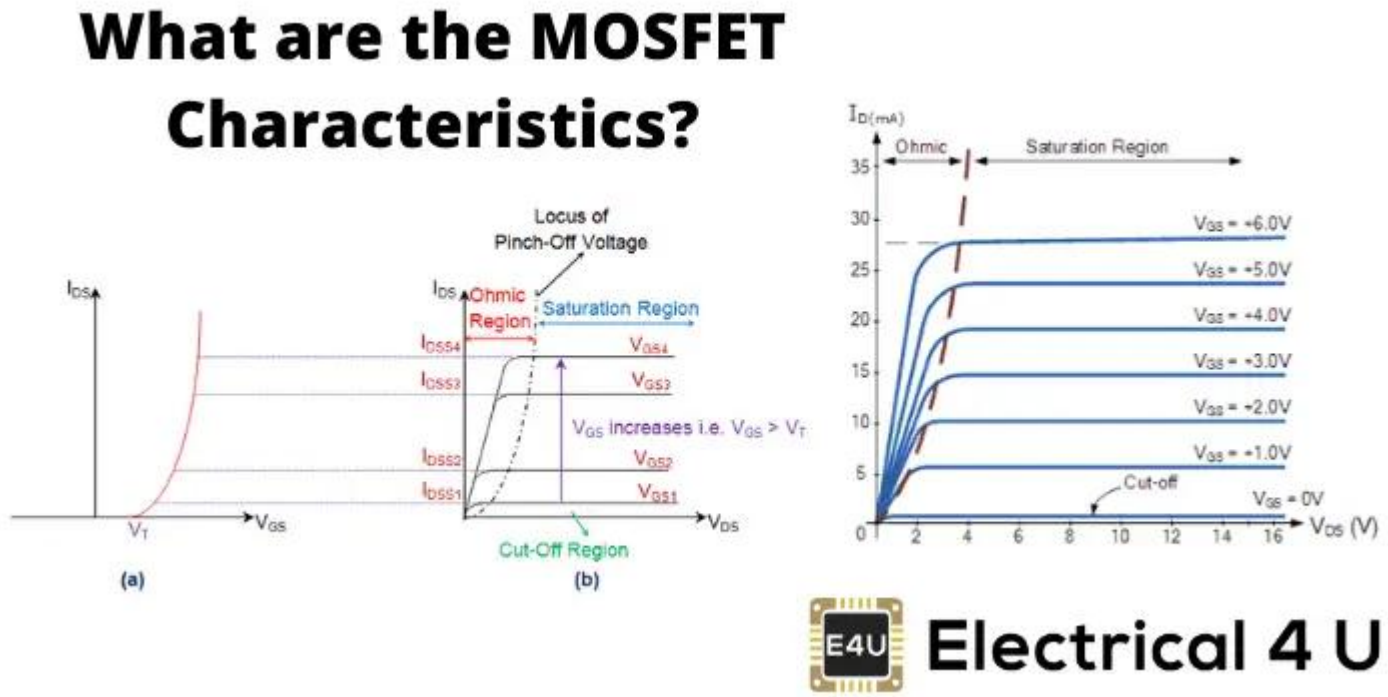
Strong channel formed

Very low resistance RDS(on)

Acts like **closed switch**

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8 V–I Characteristics of Power MOSFET



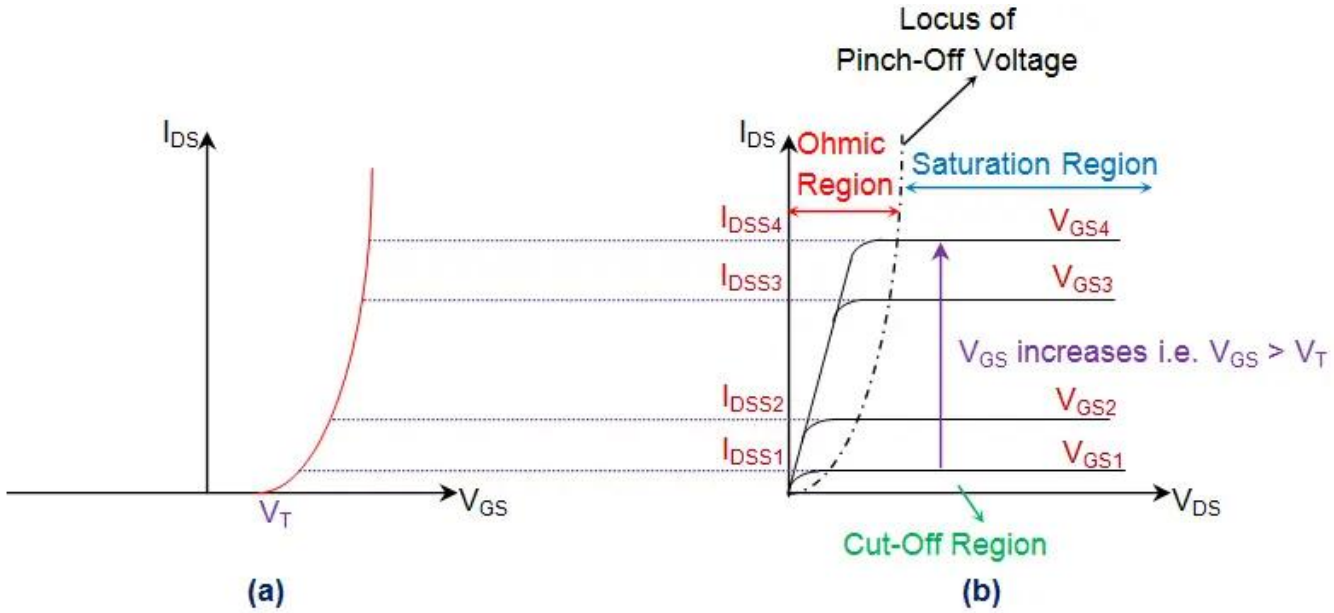


Figure 1 *n*-Channel Enhancement type MOSFET (a) Transfer Characteristics (b) Output Characteristics

Output Characteristics (IDS vs VDS)

Cut-off region

Ohmic (linear) region → switching ON

Saturation region

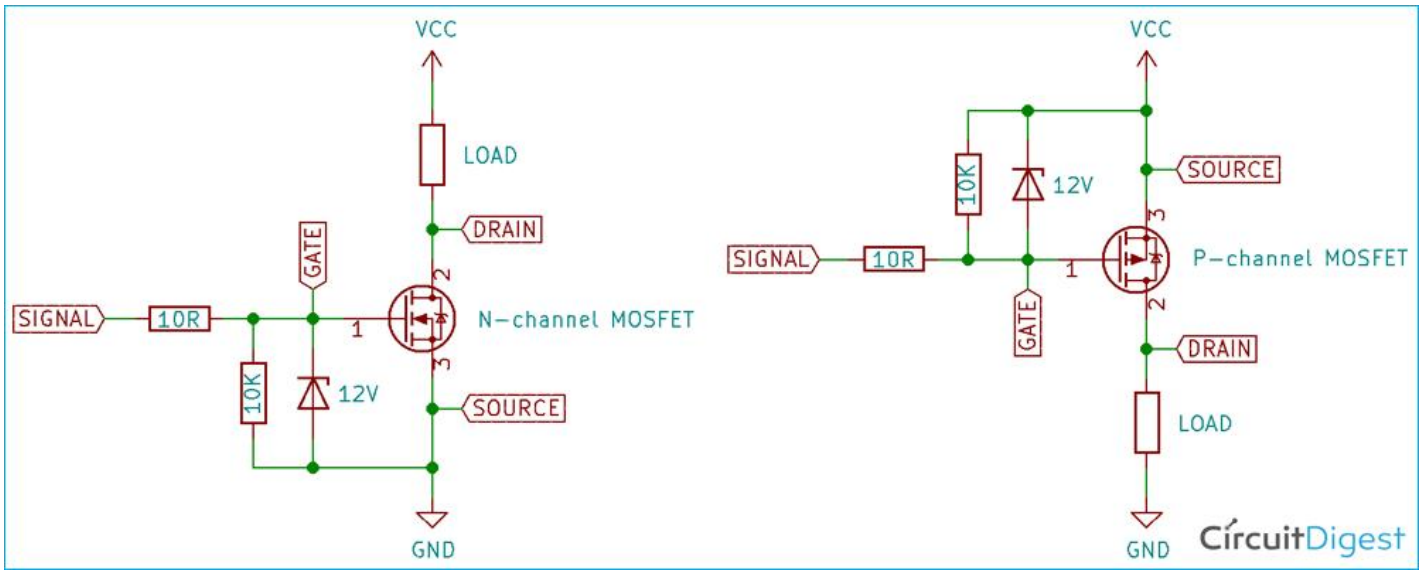
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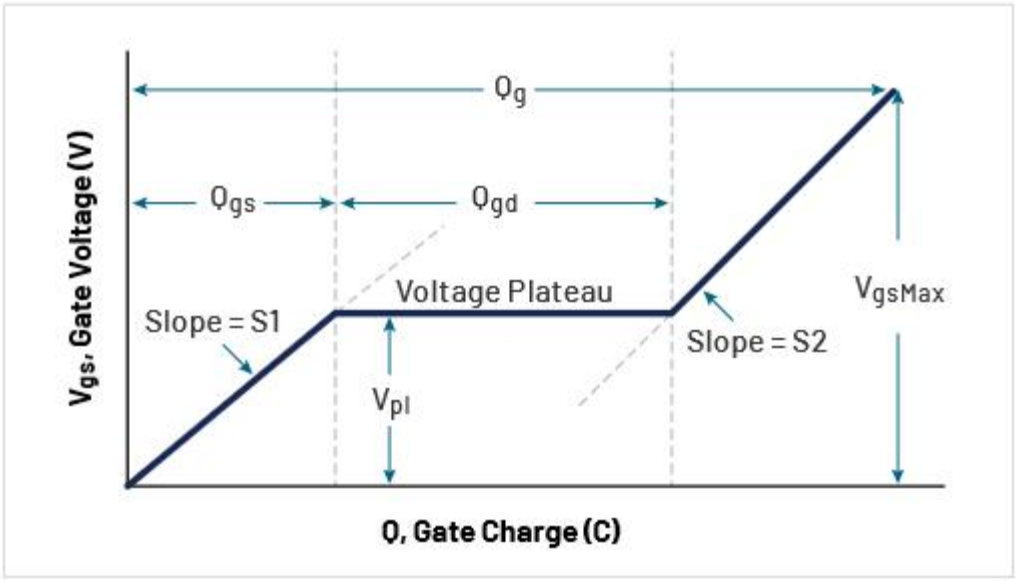
Transfer Characteristics (IDS vs VGS)

Shows gate control capability

Defines **threshold voltage Vth**

9 Switching Operation of Power MOSFET (STEP-BY-STEP)





Turn-ON process

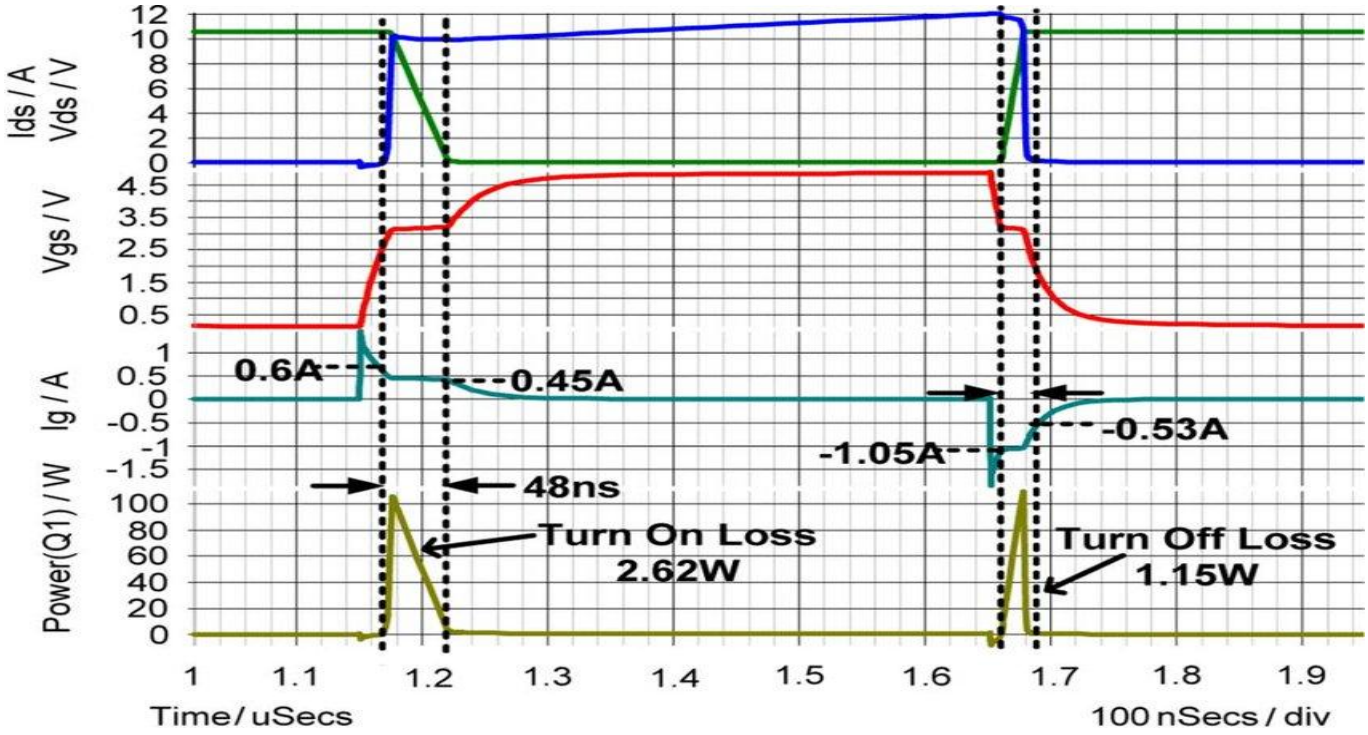
1. 1.Gate voltage applied
2. Gate capacitance charges
3. Channel forms
4. Drain current rises
5. VDS drops

Turn-OFF process

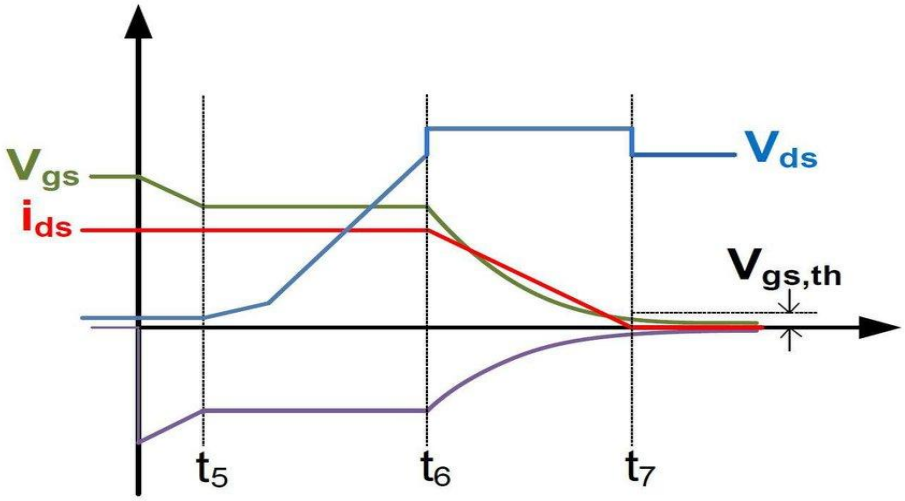
1. Gate voltage removed
2. Gate capacitance discharges
3. Channel collapses
4. Drain current falls
5. VDS rises

✓ **No charge storage → very fast switching**

Switching Waveforms (EXAM FAVORITE)



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Waveforms involved

Gate-Source voltage (VGS)

Drain current (ID)

Drain-Source voltage (VDS)

Key times:

Turn-on delay

Rise time

Turn-off delay

Fall time

Uses of Power MOSFET (2 EXAMPLES)

Example 1: SMPS

High-frequency switching

High efficiency

Example 2: DC-DC Converters

Fast switching

Low losses

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Switch Basics

What is a power electronic switch?

Device with only **two states**

ON → low resistance

OFF → high resistance

Why MOSFET is a perfect switch

- ✧ Voltage controlled
- ✧ No control current
- ✧ Fully controllable ON & OFF
- ✧ Very fast

Advantages & Limitations

✅ Advantages

- ✧ High switching speed
- ✧ Easy gate drive
- ✧ High efficiency
- ✧ No secondary breakdown

❌ Limitations

- ✧ High RDS(on) at high voltage
- ✧ Sensitive to static electricity
- ✧ Limited very-high-voltage capability

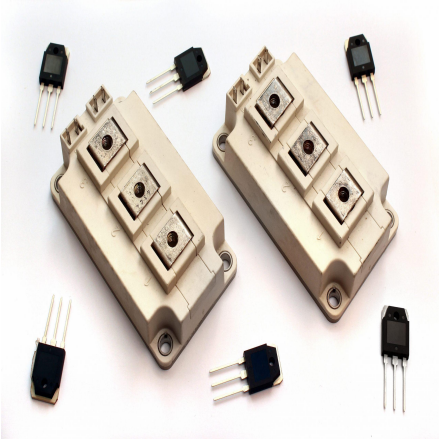
FINAL EXAM SUMMARY (WRITE THIS)

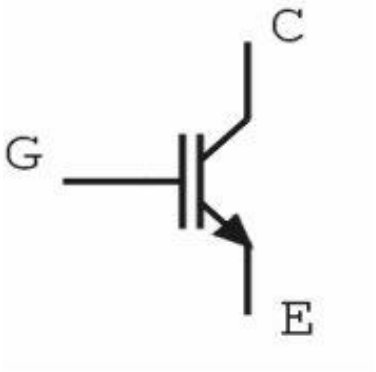
- ✧ Power MOSFET is a **voltage-controlled unipolar device**
- ✧ Gate insulated by oxide layer
- ✧ Channel formed by electric field
- ✧ Used as **high-speed power switch**
- ✧ Preferred for **high-frequency applications**

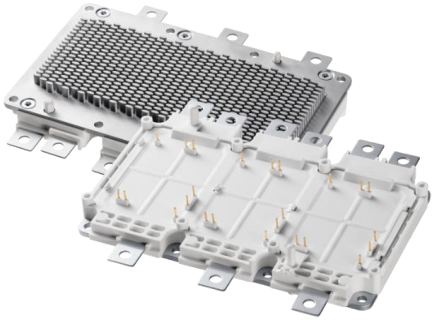
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IGBT – Insulated Gate Bipolar Transistor

1 What is an IGBT? (START HERE – BIG PICTURE)







Definition

IGBT is a **voltage-controlled power semiconductor device**

Combines:

MOSFET → easy gate control

BJT → low conduction loss at high voltage

Used where **high voltage + high current** are required

One-line identity

IGBT = MOSFET at input + BJT at output

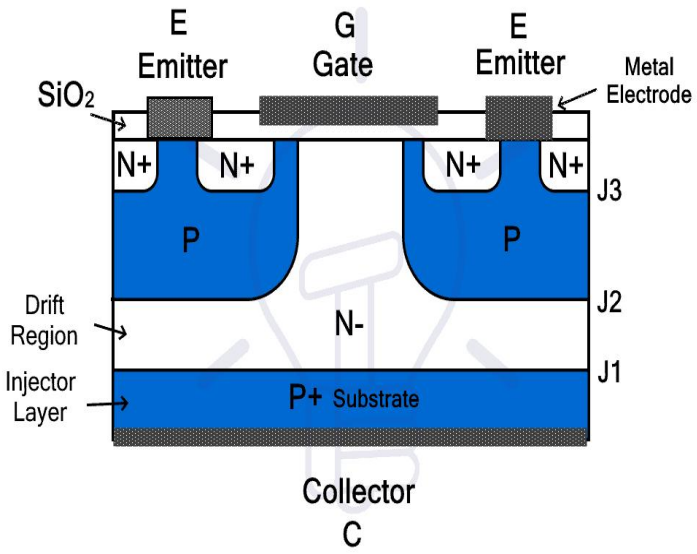
2 Why IGBT Exists (VERY IMPORTANT CONCEPT)

Device	Strength	Weakness
BJT	Low ON loss	Needs base current, slow
MOSFET	Fast switching	High RDS(on) at high voltage
IGBT	Low ON loss + easy drive	Slightly slower turn-off

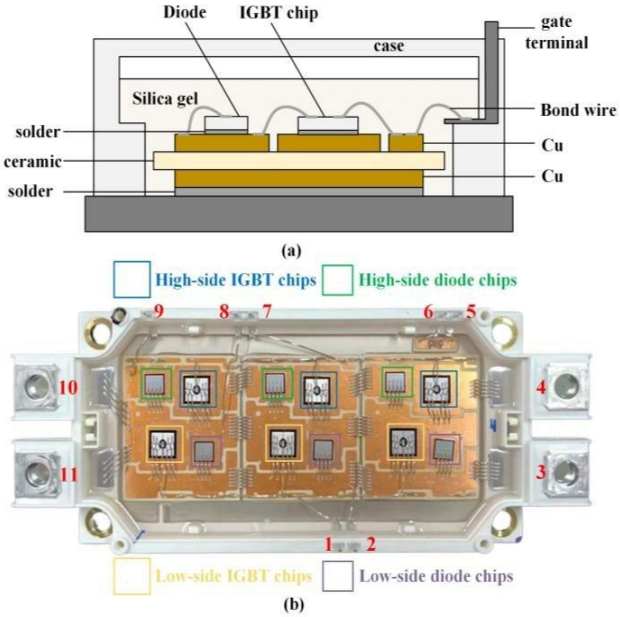
IGBT fills the **gap between MOSFET and BJT**

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3 Construction of IGBT (FOUNDATION)



Structure of IGBT



Terminals

Gate (G) – control terminal (voltage-controlled)

Collector (C) – power terminal

Emitter (E) – reference terminal

Internal layers

P^+ Collector

N^- Drift region

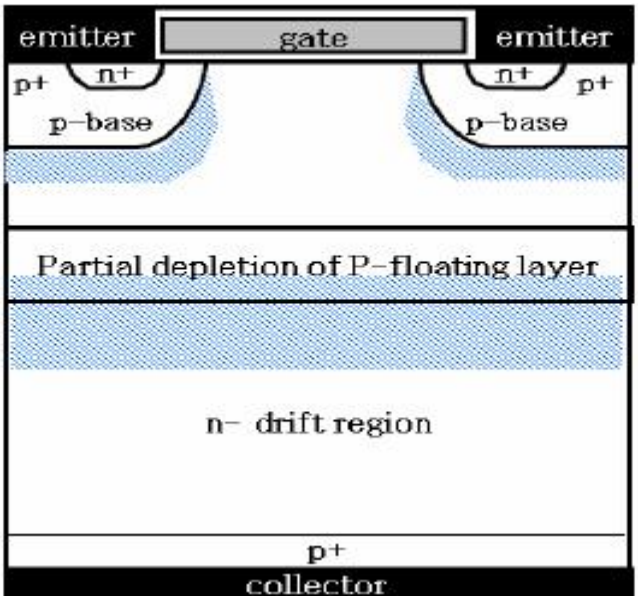
P-body

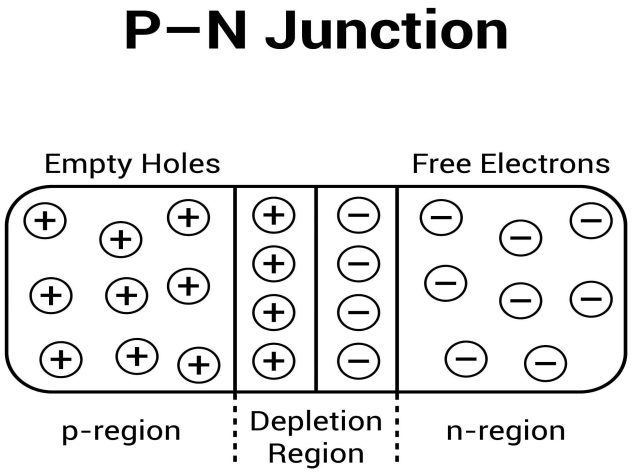
N^+ Emitter

Gate oxide (SiO_2)

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4 Junctions & Depletion Layer in IGBT





PN Junctions inside IGBT

P^+ collector – N^- drift

P-body – N^+ emitter

Depletion layer

Mainly in **N^- drift region**

Supports **high blocking voltage**

Widens during OFF state

Drift region thickness = voltage rating

5 Operating Principle of IGBT

Gate voltage controls a MOS channel, which activates a bipolar conduction path

Key idea

Gate voltage creates **MOS channel**

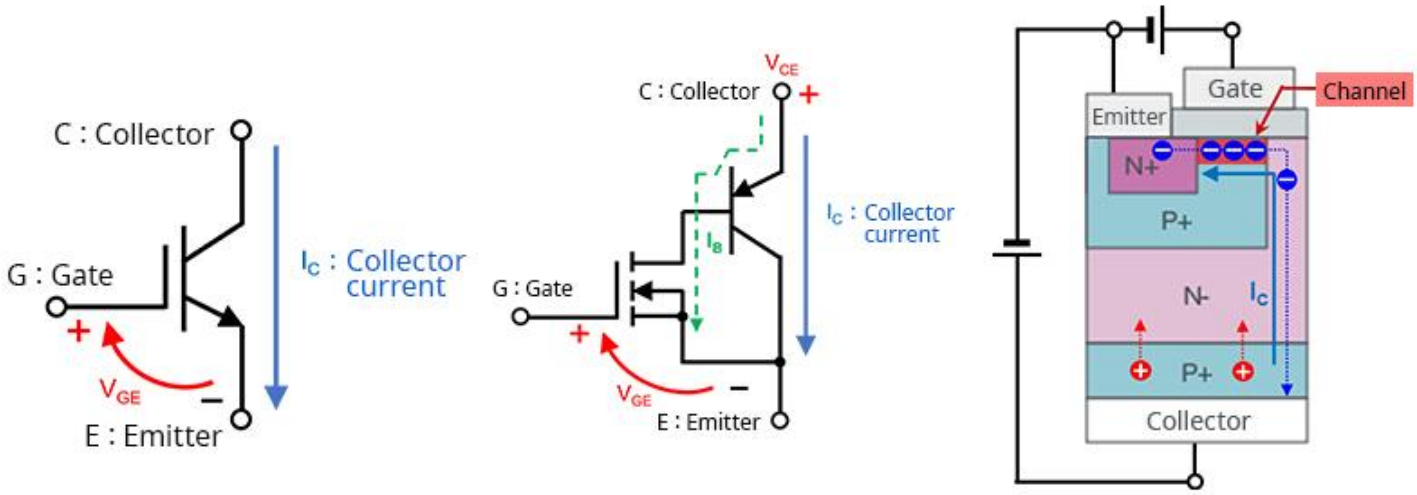
Channel allows electrons to flow

This turns ON internal **PNP transistor**

Result: **Bipolar conduction with low ON loss**

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6 Working of IGBT (STEP-BY-STEP)



Step-1: Gate voltage applied

VGE > threshold (~5–7 V)

MOS channel forms

Step-2: Electron flow starts

Electrons flow from emitter to drift region

Step-3: Bipolar action starts

Holes injected from collector

Conductivity modulation occurs

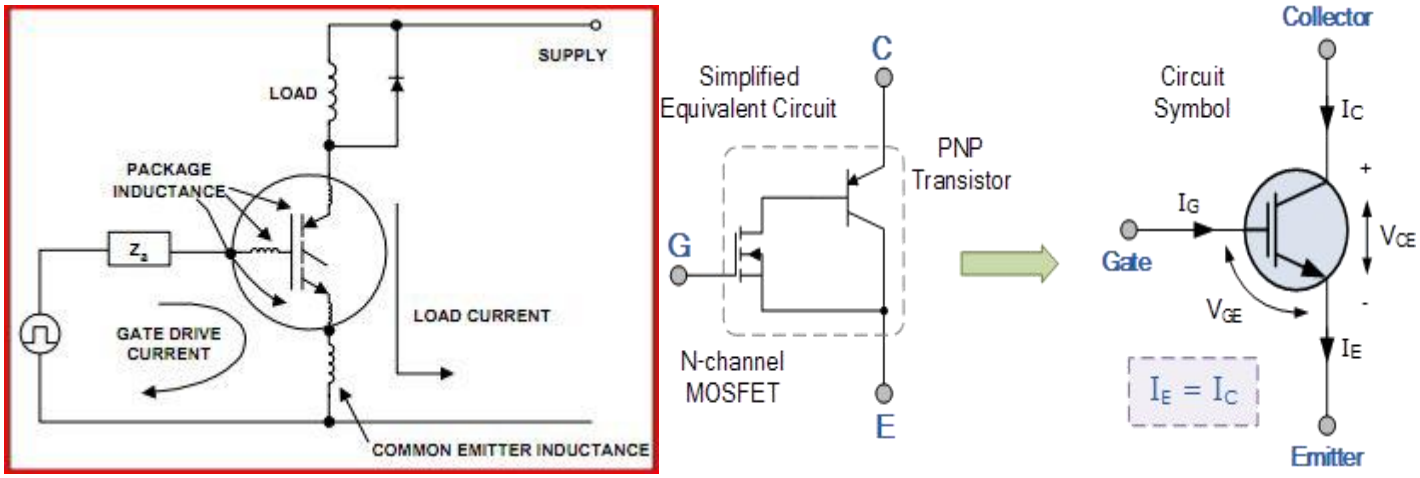
Step-4: High current conduction

Low VCE(sat)

Very efficient at high voltage

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7 IGBT as a SWITCH (MOST IMPORTANT)



OFF State

VGE = 0

No channel

Depletion region wide

IC ≈ 0

Acts like **open switch**

ON State

VGE = 10–15 V

Channel formed

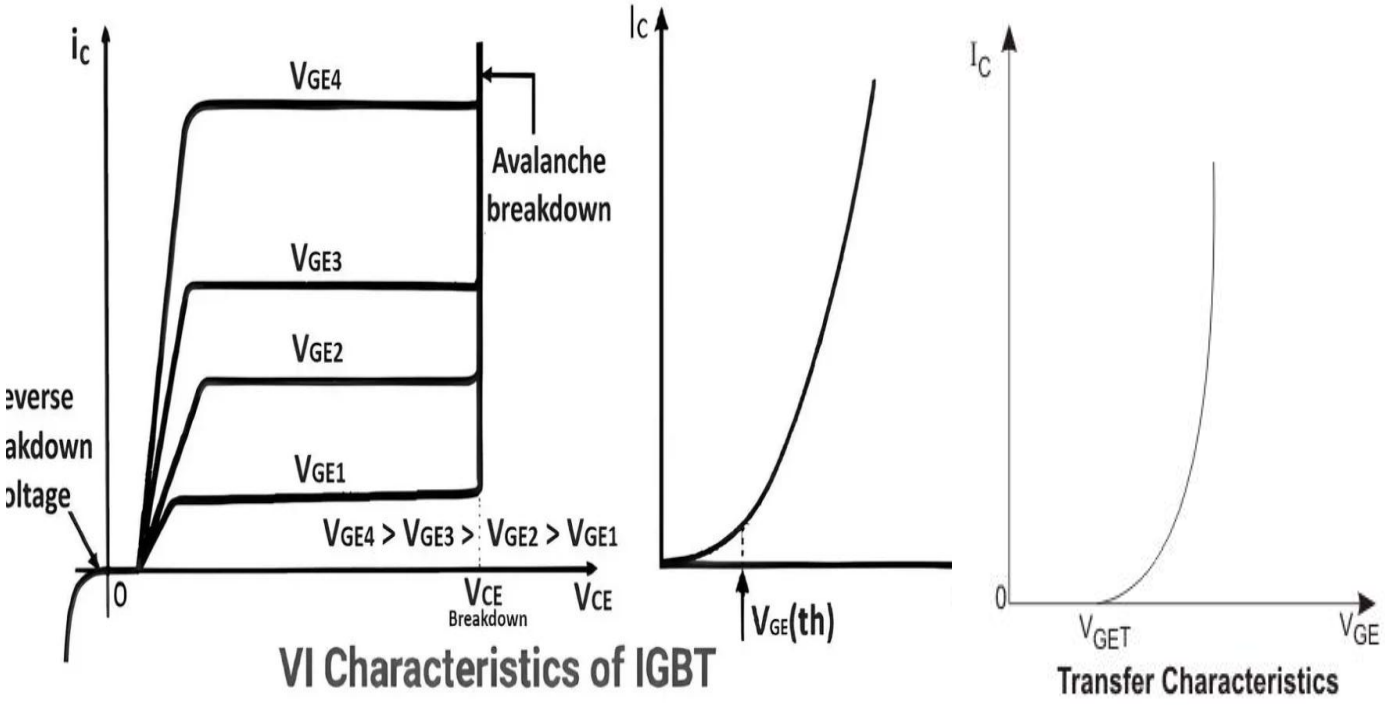
Bipolar conduction

VCE(sat) ≈ 1.5–3 V

Acts like **closed switch**

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8 V–I Characteristics of IGBT



Output Characteristics (IC vs VCE)

Similar to BJT

Flat curve → current controlled by gate

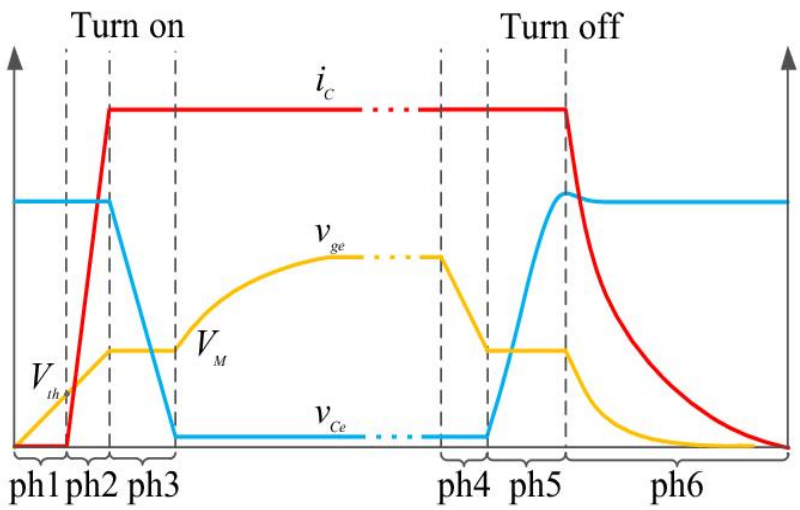
Transfer Characteristics (IC vs VGE)

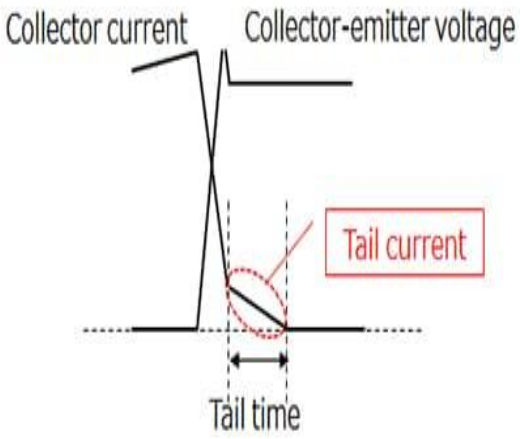
Shows voltage-controlled nature

Higher VGE → higher IC

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9 Switching Operation of IGBT (STEP-BY-STEP)





Turn-ON

Gate voltage applied

Gate capacitance charges

MOS channel forms

Bipolar conduction begins

IC rises, VCE falls

Turn-OFF (CRITICAL POINT)

Gate voltage removed

MOS channel disappears

Stored charge remains in drift region

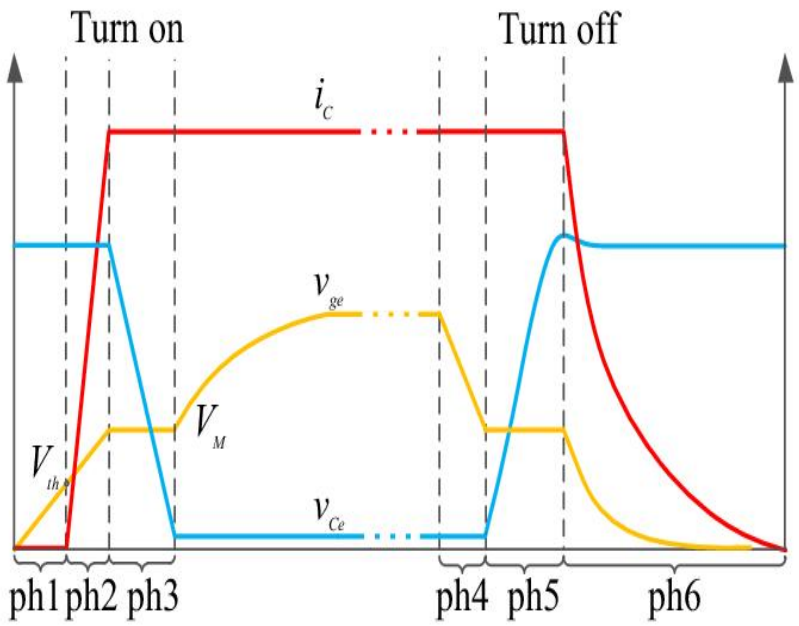
Tail current flows

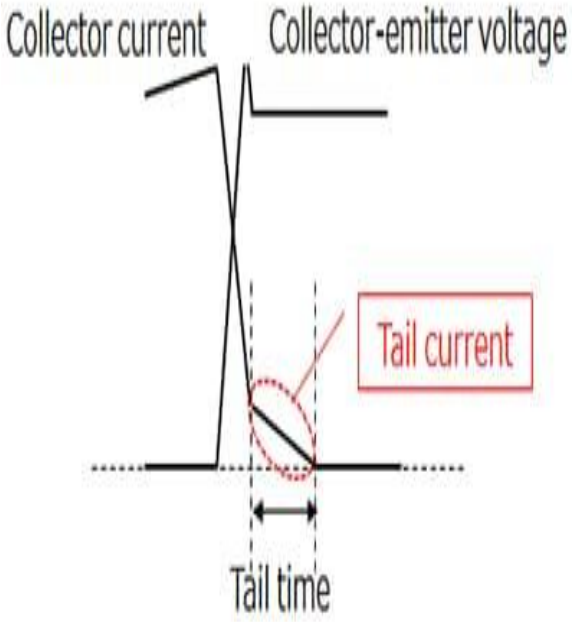
IC slowly falls → OFF

Tail current = main drawback of IGBT

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Switching Waveforms





Waveforms

Gate-Emitter voltage (VGE)

Collector current (IC)

Collector-Emitter voltage (VCE)

Key observation:

Tail current during turn-OFF

Higher switching loss than MOSFET

Uses of IGBT (2 STRONG EXAMPLES)

Example 1: EV Traction Inverter

High voltage DC

Large motor current

IGBT gives efficiency + reliability

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Example 2: Industrial Motor Drives

Medium switching frequency

High power handling

Switch Basics

What is a power switch?

ON → low impedance

OFF → high impedance

Why IGBT is a fully controlled switch

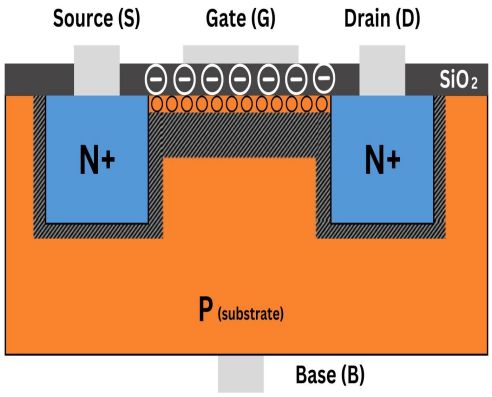
Gate controls turn-ON

Gate controls turn-OFF

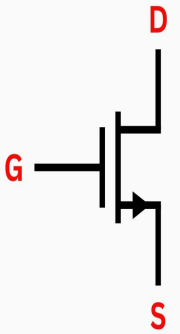
No external commutation needed

TYPES OF MOSFET (AS REQUESTED)

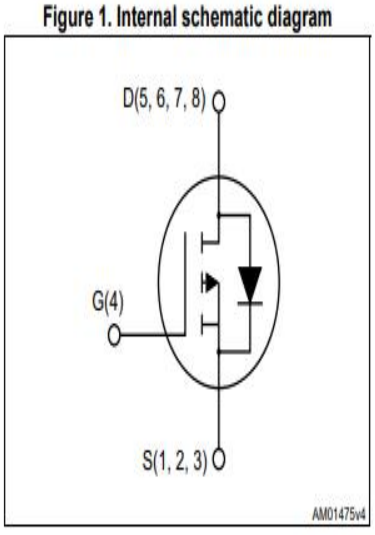
N-Channel MOSFET



N-Channel MOSFET: Structure



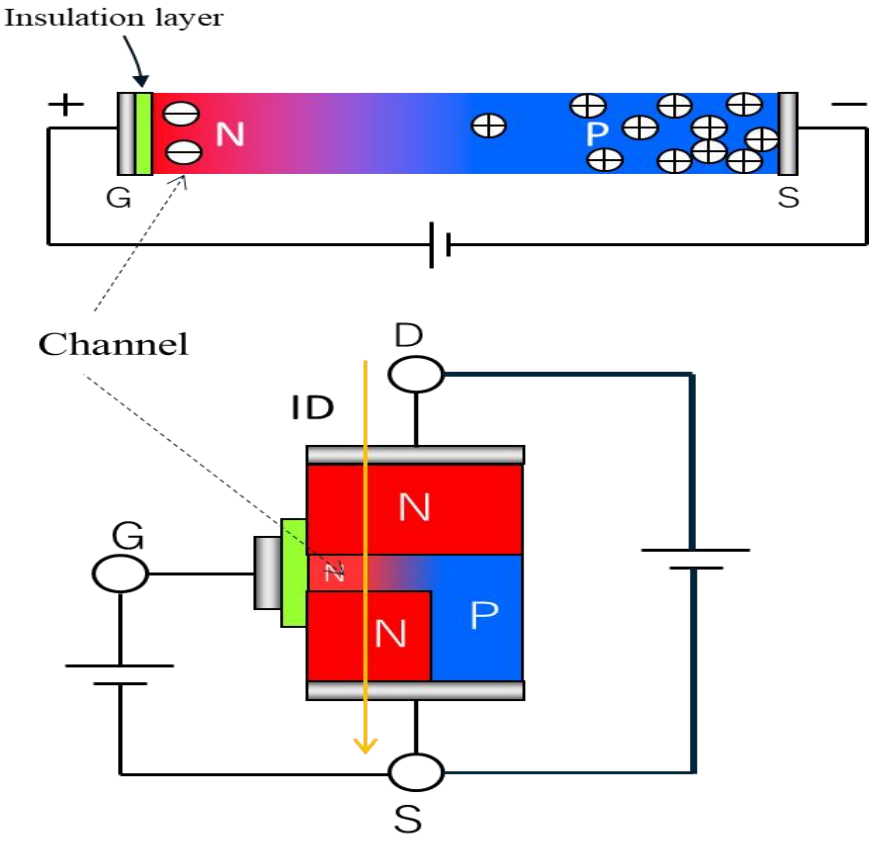
Symbol

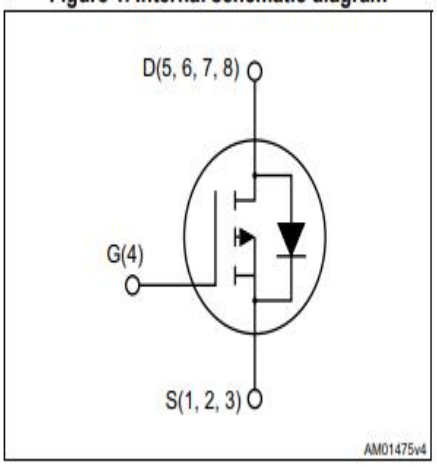


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- ✧ Electron conduction
- ✧ Fastest
- ✧ Lowest RDS(on)
- ✧ Most widely used

P-Channel MOSFET





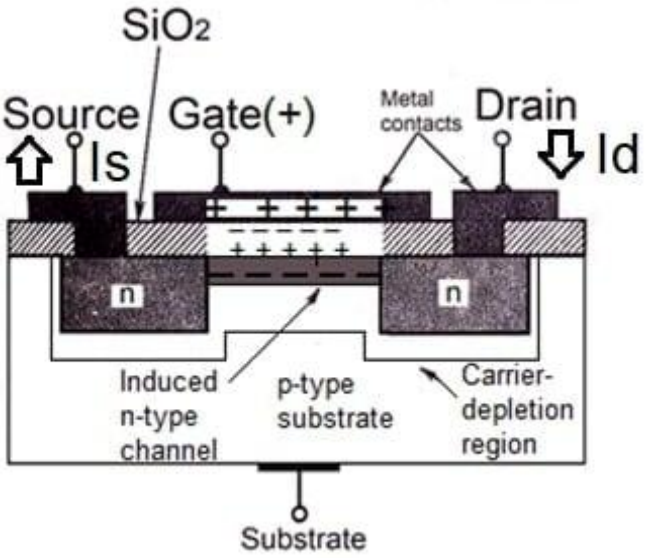
Hole conduction

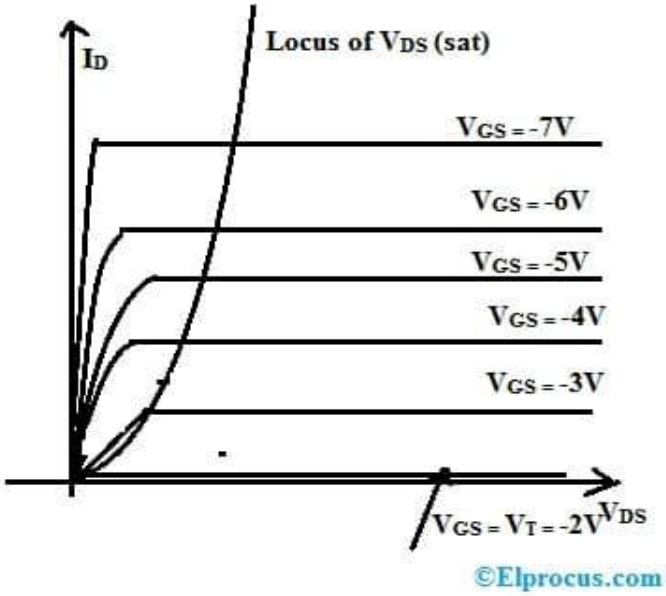
Slower

Used for high-side switching

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Enhancement-Mode MOSFET



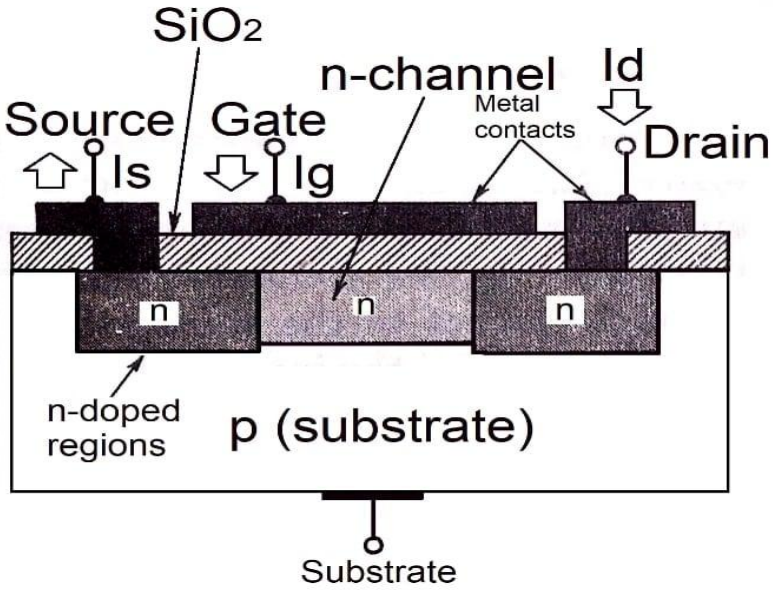


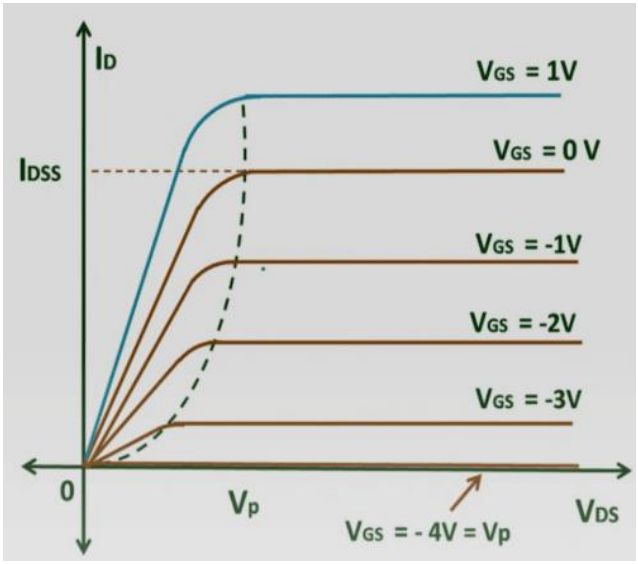
Normally OFF

Needs VGS > Vth

Used in power electronics

Depletion-Mode MOSFET





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Normally ON

Rare in power circuits

FINAL COMPARISON (PRO LEVEL INSIGHT)

Device	Speed	Voltage	Control
BJT	Slow	Medium	Current
MOSFET	Fast	Low–Medium	Voltage
IGBT	Medium	High	Voltage

FINAL TAKEAWAY (WRITE THIS)

IGBT = **MOS gate + BJT conduction**

Used for **high-power switching**

Low conduction loss

Moderate switching speed

Dominant device in **EVs & industrial drives**

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